Lab Class Task

Submitted by

Saba Ashraf

Roll no: 24

CS Section 'A'

Submitted to

Sir Asad Majeed

Task - 1

Run the given program and observe their output

```
#include <iostream>
using namespace std;
int main ()
{
```

```
Char letter;
letter = 'A';
cout << letter <<"\n";
. letter = 'B';
 cout << letter<<"\n";
  cin >> letter;
  return 0;
}
```

Output

A
B

```
# include<iostream>
using namespace. std;
 int main ( )
{
    char letter ;
      letter = 65;
    cout << letter <<"\n";
      letter = 66;
      cout << letter << "\n";
       cin >> letter ;
     return 0;
}
```

Output

A
B

```
#include <iostream>
using namespace std;
int main ()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

output

65
66

Task - 2

Take two integers as input and apply arthematic operations on them (+, -, *, /) as print them on screen.

```
#include <iostream>
using namespace std;
int main()
{
    int a= 10;
    int b= 2;
  cout << a+b << endl;
  cout << a-b << endl;
  cout << a*b << endl;
  cout << a/b << endl;
}
```

Output

12
8
20
5

Task - 3

write the output of the following code.

```
#include <iostream>
using namespace std;
int main()
{
    int val1;
    int val2;
    int val3;
    cout<<"Please enter your 3 number:";
    cin>>val1>>val2>>val3;
    cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
    cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
    return 0;
}
```

Output

Please enter your 3 number: 1 3 8

1
3
8
8
3
1

Task - 4

Write a C++ Program to calculate the distance between the two points.
Note: $x1, y1, x2, y2$ are all double values.

```
#include<iostream>
#include<cmath>
using namespace std;
 int main()
 {
     double x1, y1, x2, y2;
      double distance;
    Cout<<"Enter x1:";
    Cin>>x1;
    Cout<<"Enter y1:";
    Cin>>y1;
    Cout<<"Enter x2:";
     Cin>>x2;
     Cout<<"Enter y2:";
     Cin>> y2;
   distance= sqrt(pow(x2-x1,2)+ pow(y2-y1,2));
    Cout<<"Distance between two points is:"<<distance<<endl;
      return 0;
}
```

Output

```
Enter x1: 1
Enter y1: 2
Enter x2: 4
Enter y2: 6
Distance between two points is 5
```

Task - 5

(1) Calculate the area of circle (area = pi x r^2)

```
#include<iostream>
using namespace std;
int main()
 {
   double r, area;
   const double pie= 3.14159;
   cout<<"Enter radius of circle:";
    cin >> r;
   area = pie * r * r;
   cout << "Area of a circle:"<< area<<endl;
     return 0;
   }
```

Output

```
Enter radius of a circle:5
Area of a circle:78.5397
```

Calculate the Area of rectangle (area = length * width)

```
# include <iostream>
using namespace std;
int main ( )
 {
      float length, width, area;

      Cout << "Enter length of rectangle :" ;
       Cin >> length ;
       Cout << " Enter width of rectangle :" ;
        cin >> width ;
       area = length * width ;
      Cout << "Area of rectangle :" << area << endl ;
        return 0 ;
 }
```

output

```
Enter length of rectangle : 4.2
Enter width of rectangle : 5.1
area of rectangle : 21.42
```

Calculate the Area of a triangle (area=b*h*0.5)

```
# include < iostream>
using name space std;
int main( )
  {
      float base, height, area;
       cout << " Enter base of triangle: ";
        cin >> base;
        cout << " Enter height of triangle:";
          cin >> height;
       area = base * height * 0.5
       cout << " Area of triangle": << area << end.l;
         return 0;
  }
```

output

Enter base of triangle: 5.6

Enter height of triangle: 8.1

Area of triangle : 22.68

Task - 6

Total Purchase A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item5 = $3.95

Write a Program that holds the prices of the fives items in five variables. Display each items Price, the subtotal of the sale, the amount of sales tax and the total. Assume the sales tax is 6%.

```
#include <iostream>
using namespace std;
int main ()
{
    float item 1 = 12.95f;
    float item 2 = 24.95f;
    float item 3 = 6.95;
    float item4 = 14.95;
    float item5 = 3.95f
```

```
float Subtotal, SalesTax , Total;
 const float Tax_Rate = 0.06f ;
Subtotal = item1 + item2+ item3+ item4+ item5;
Sales Tax = Subtotal * Tax_Rate ;
 Total = Subtotal + SalesTax ;
Cout<<"Price of item1: $"<<item1<<endl;
Cout<<"Price of item2: $"<<item2<<endl;
cout<<"Price of item3: $"<<item3<<endl;
cout<<"Price of item4: $"<<item4<<endl;
cout<<"Price of item5: $"<<item5<<endl;

 cout<<"Subtotal of the sale: $"<<Subtotal<<endl;
cout<<"Amount of SaleTax (6%): $"<<Sale tax<<endl;
cout<<" Total : $"<<Total<<endl;
 return 0;
}
```

output

```
Price of item1 : $12.95
Price of item2 : $24.95
Price of item3: $6.95
Price of item4: $14.95
Price of item5: $ 3.95
Subtotal of the sale: 63.75
Amount of SalesTax : 3.825
 Total : $ 67.575
```

Bonus Task :

Break Number:
Write a program that inputs a five-digit integer, separates the integer into its individual digits and print the digits seprated from one another by three spaces.

```
# include <iostream>
using namespace std;
int main ()
 {
      int num;
     Cout << "Enter 5 digit number: ";
      cin >> num;
    int d1 = num / 10000;
    int d2 = (num/1000) % 10;
    int d3 = (num/100) % 10;
    int d4 = (num/10) % 10;
     int d5 =  num % 10;
Cout<<d1<<"   "<<d2<<"   "<<d3<<"   "<<d4<<"   "<<d5;
   return 0;
 }
```

Output:

```
Enter 5 digit number: 23456
2   3   4   5   6
```

Home Tasks

Task 1:

Exchange write a program that reads in two variables and swaps the value of these variables. For example if a variable 'var 1' contains 10 and variable 'var 2' contains 20 after the swap operation the variable 'var 1' contains 20 and variable 'var 2' contain 10.

```
#include <iostream>
using namespace std;
int main ()
{
    int var1 = 10;
    int var2 = 20;
    int tem;
    temp = var1;
    var1 = var2;
    var2 = tem;
cout << " var 1 =" << var1 << endl;
cout << " var 2 =" << var2 << endl;
return 0;
}
```

output

var1 = 20

var2 = 10

Task 2

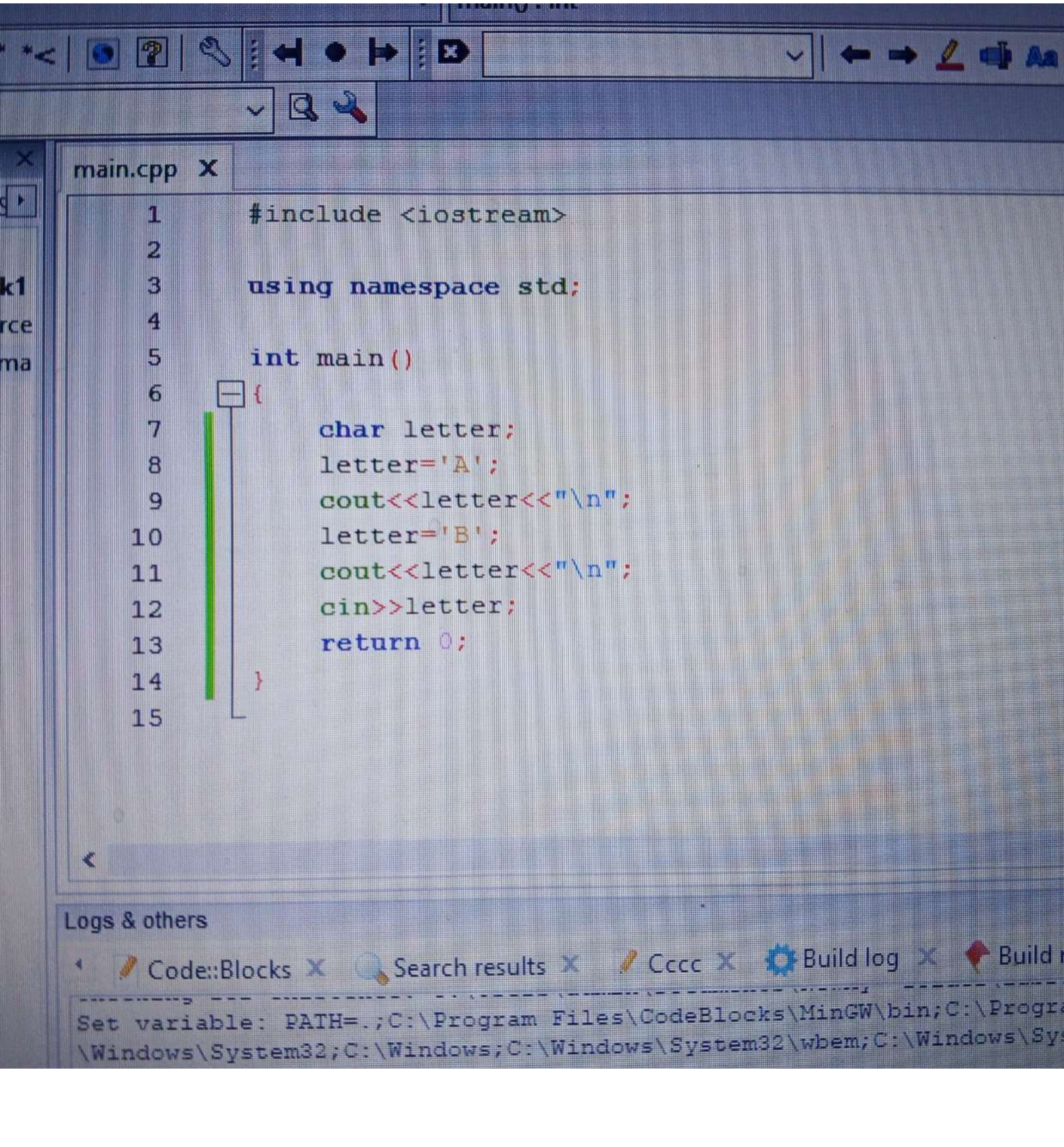
Make Upper Write a program that prompts the user to enter an alphabet in small caps (a, b, ... z) and display the entered alphabet into its upper caps (A, B, ... Z)

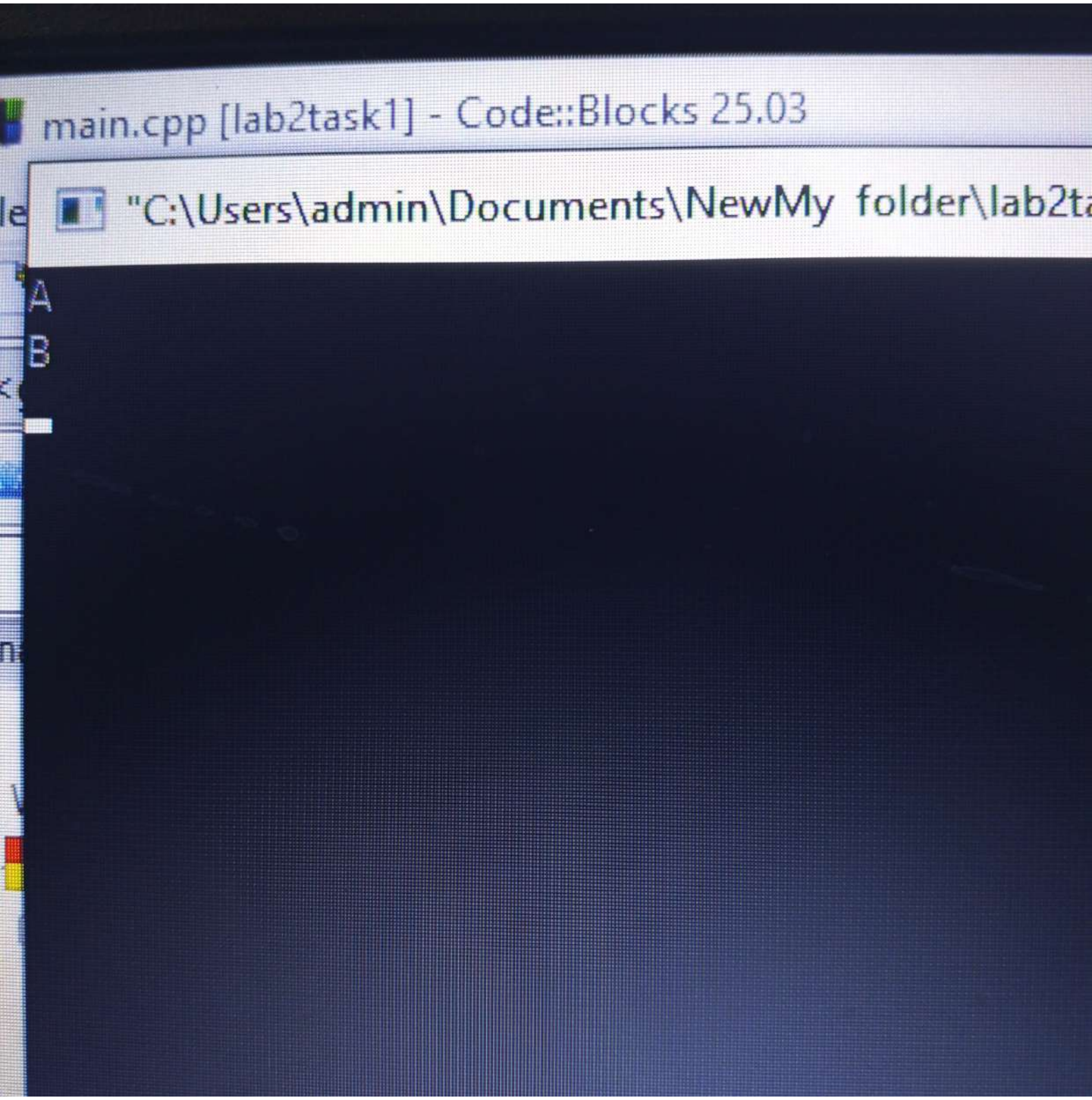
```
#include<iostream>
using namespace std;
 int main ()
{
      char  small, capital;
       cout << " Enter small letter :";
        Cin >> small;
        Capital = small - 32;
     Cout<< " Capital letter : "<< capital;
      return 0;
}
```

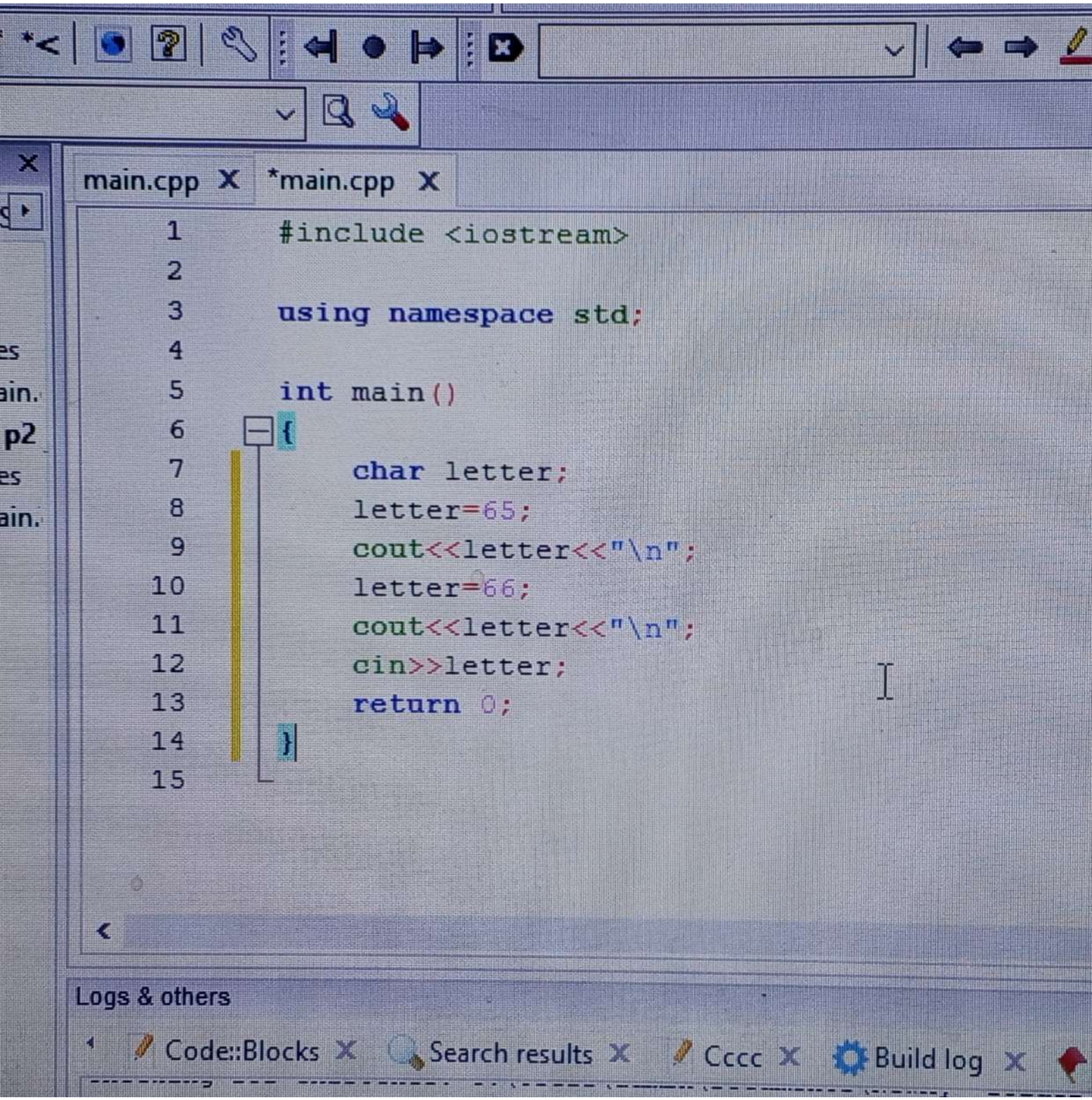
Output

Enter small letter : b

Capital letter : B


main.cpp
1 #include <iostream>
2
3 using namespace std;
4
5 int main()
6 {
7 char letter;
8 letter='A';
9 cout<<letter<<"\n";
10 letter='B';
11 cout<<letter<<"\n";
12 cin>>letter;
13 return 0;
14 }
15
Logs & others
Code::Blocks
Search results
Cccc
Build log
Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\bin;C:\Progr
\Windows\System32;C:\Windows;C:\Windows\System32\wbem;C:\Windows\Sy


main.cpp [lab2task1] - Code::Blocks 25.03
"C:\Users\admin\Documents\NewMy folder\lab2ta
A
B



main.cpp | *main.cpp

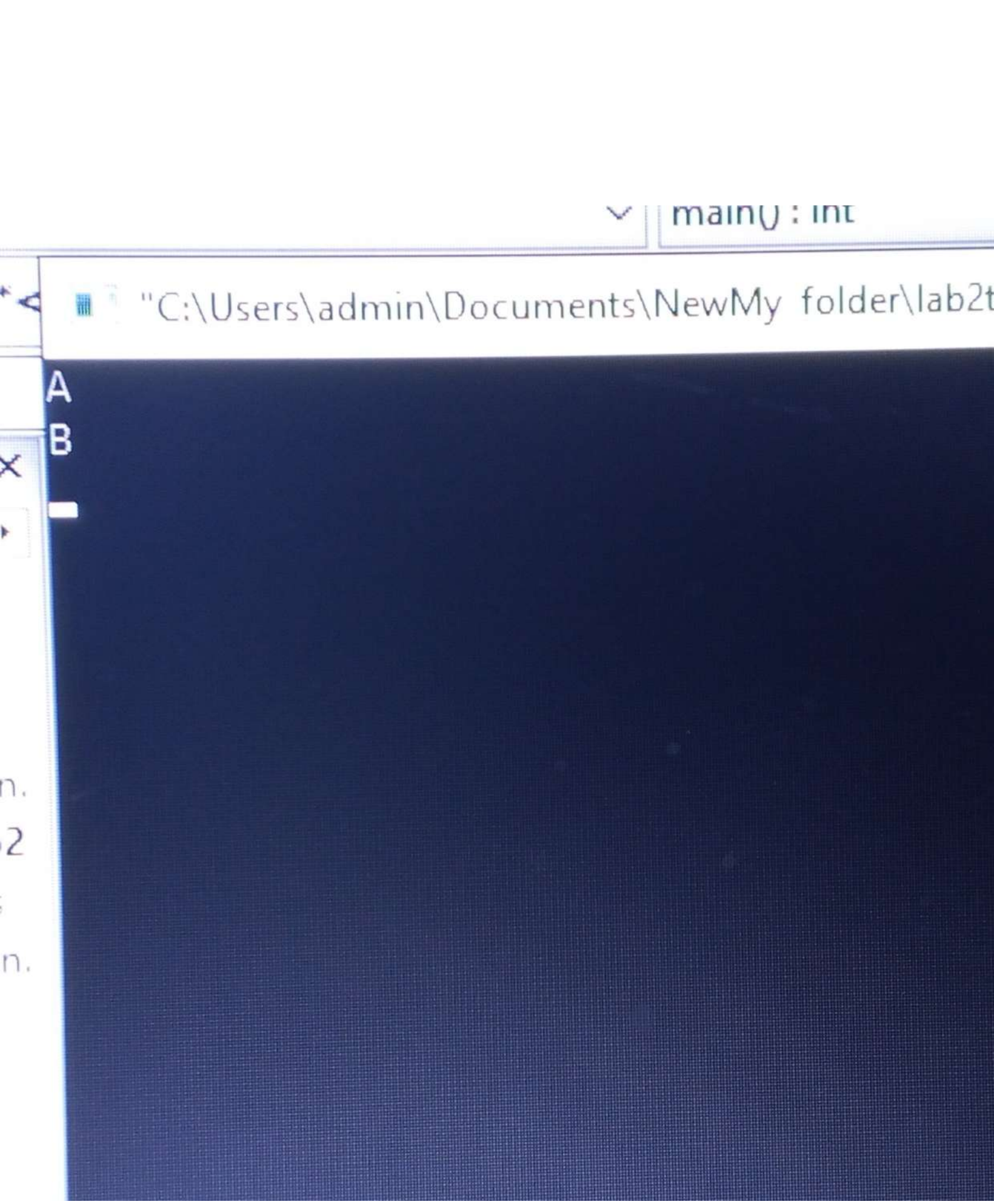
```
#include <iostream>

using namespace std;

int main()
{
    char letter;
    letter=65;
    cout<<letter<<"\n";
    letter=66;
    cout<<letter<<"\n";
    cin>>letter;
    return 0;
}
```

Logs & others

Code::Blocks | Search results | Cccc | Build log


main() : int
"C:\Users\admin\Documents\NewMy folder\lab2t
A
B

main.cpp | main.cpp | *main.cpp

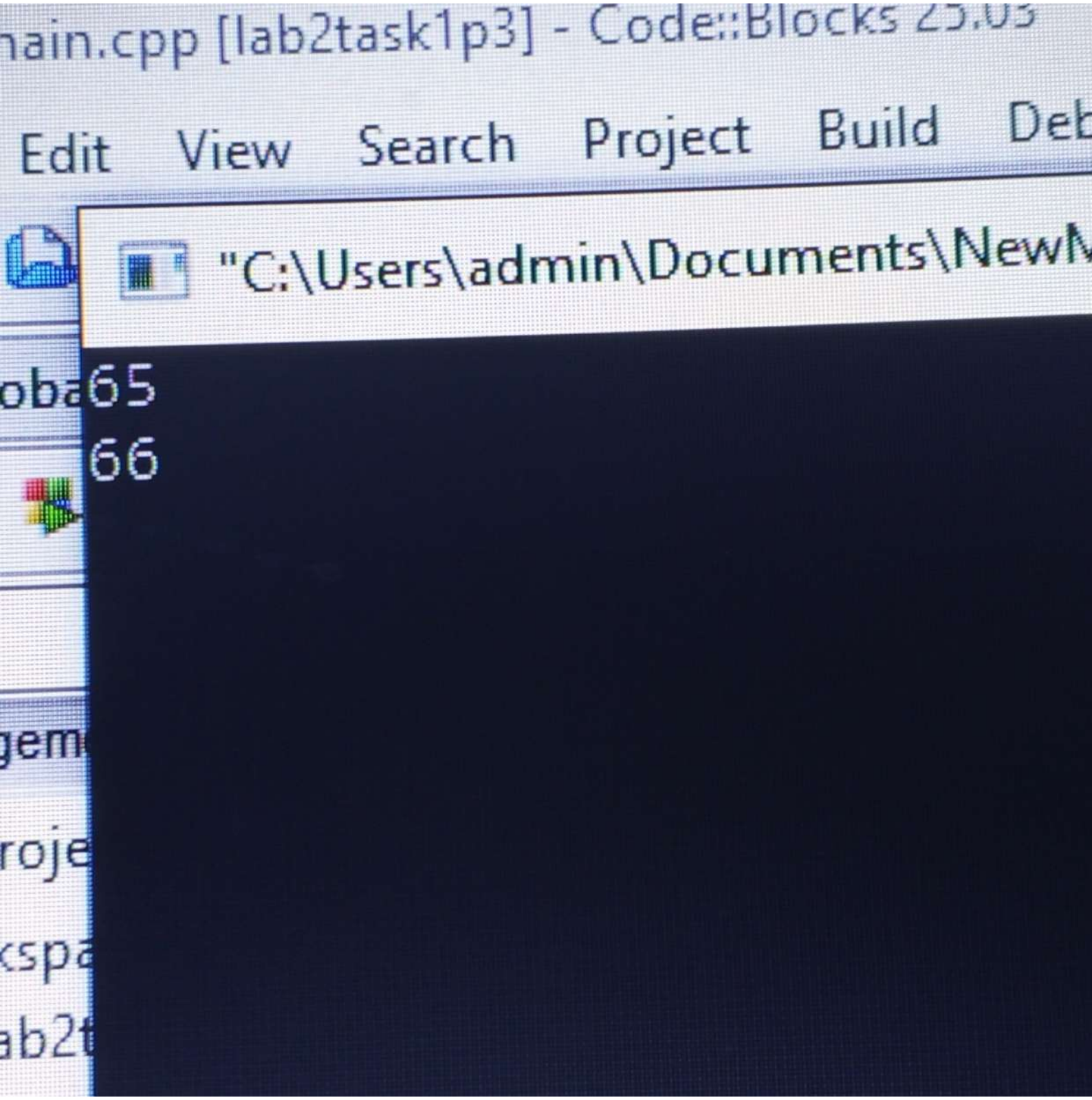
```
1   #include <iostream>
2
3   using namespace std;
4
5   int main()
6   {
7       int letter;
8       letter='A';
9       cout<<letter<<"\n";
10      latter='B';
11      cout<<letter<<"\n";
12      cin>>letter;
13      return 0;
14  }
15
```

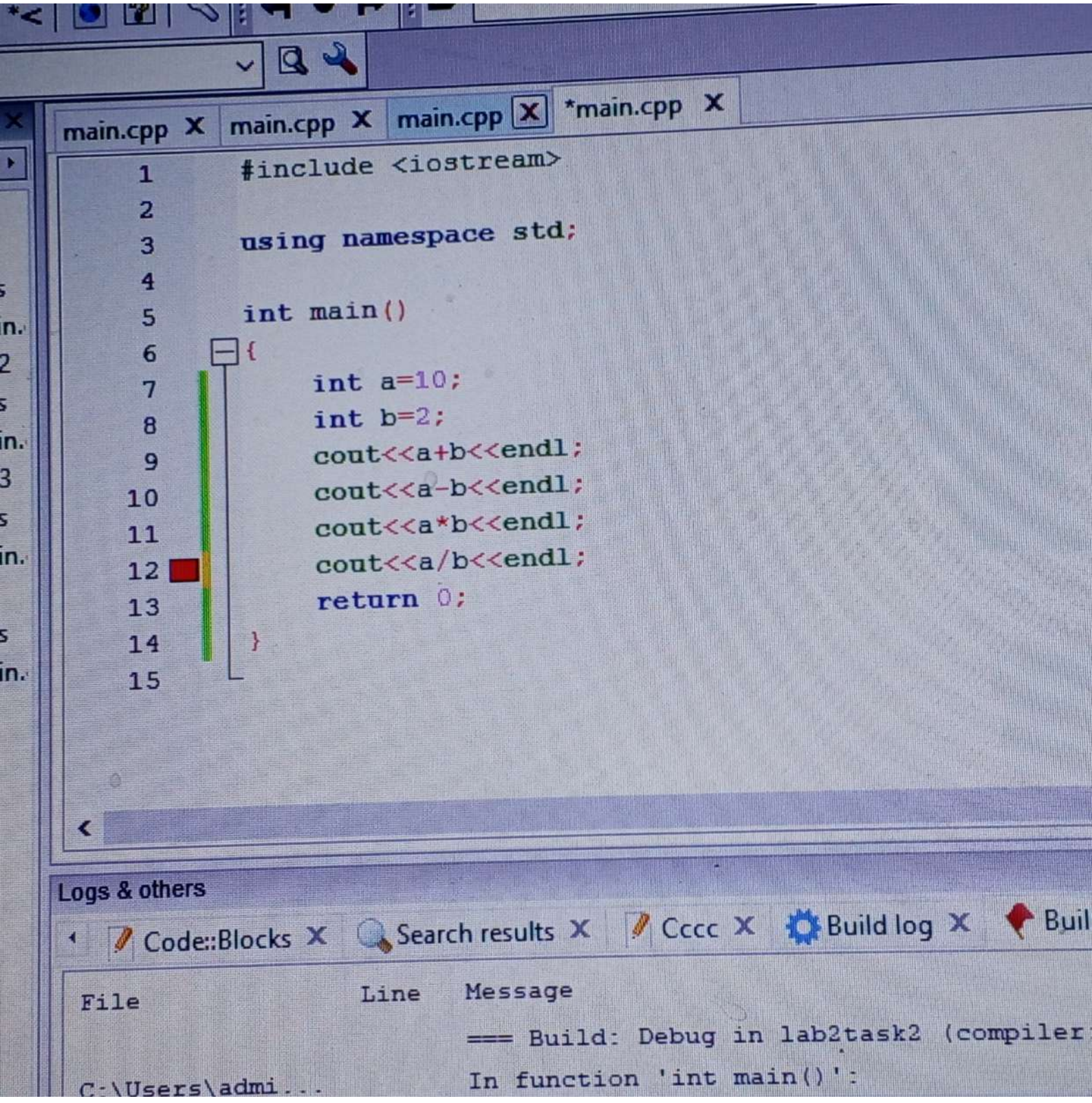
Logs & others

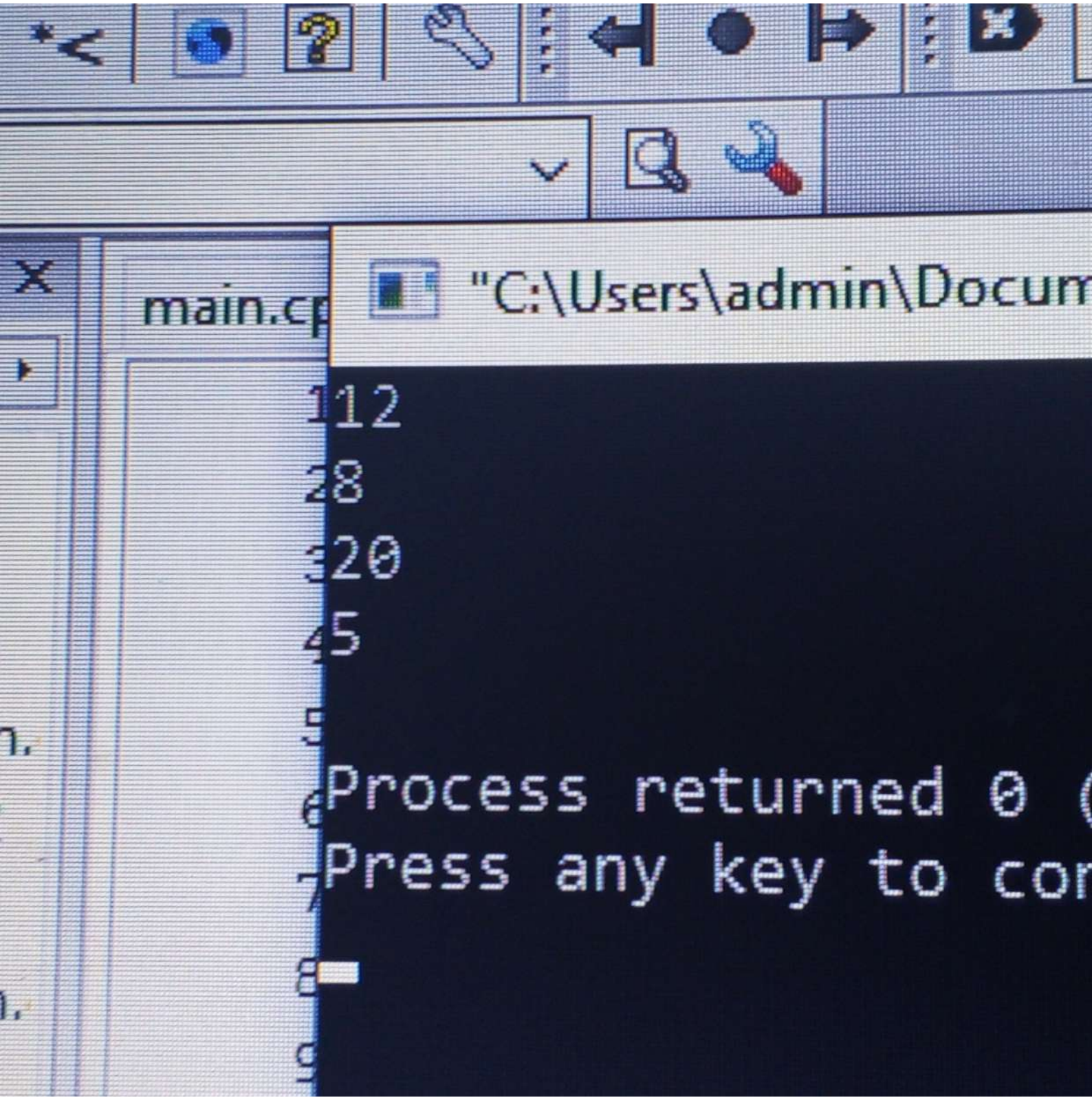
Code::Blocks | Search results | Cccc | Build log | Build messag

```
Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Fil
\Windows\System32;C:\Windows;C:\Windows\System32\wbem;C:\Windows\System32
\Microsoft\WindowsApps
Executing: "C:\Program Files\CodeBlocks/cb_console_runner.exe" "C:\Users\
\admin\Documents\NewMy  folder\lab2task1p2\.)
```

```
-1073741510 (0 minute(s), 49 second(s))
```


ain.cpp [lab2task1p3] - Code::Blocks 25.03
Edit View Search Project Build Deb
"C:\Users\admin\Documents\NewM
65
66


main.cpp
main.cpp
main.cpp
*main.cpp
1 #include <iostream>
2
3 using namespace std;
4
5 int main()
6 {
7 int a=10;
8 int b=2;
9 cout<<a+b<<endl;
10 cout<<a-b<<endl;
11 cout<<a*b<<endl;
12 cout<<a/b<<endl;
13 return 0;
14 }
15
Logs & others
Code::Blocks
Search results
Cccc
Build log
File
Line
Message
=== Build: Debug in lab2task2 (compiler
In function 'int main()':

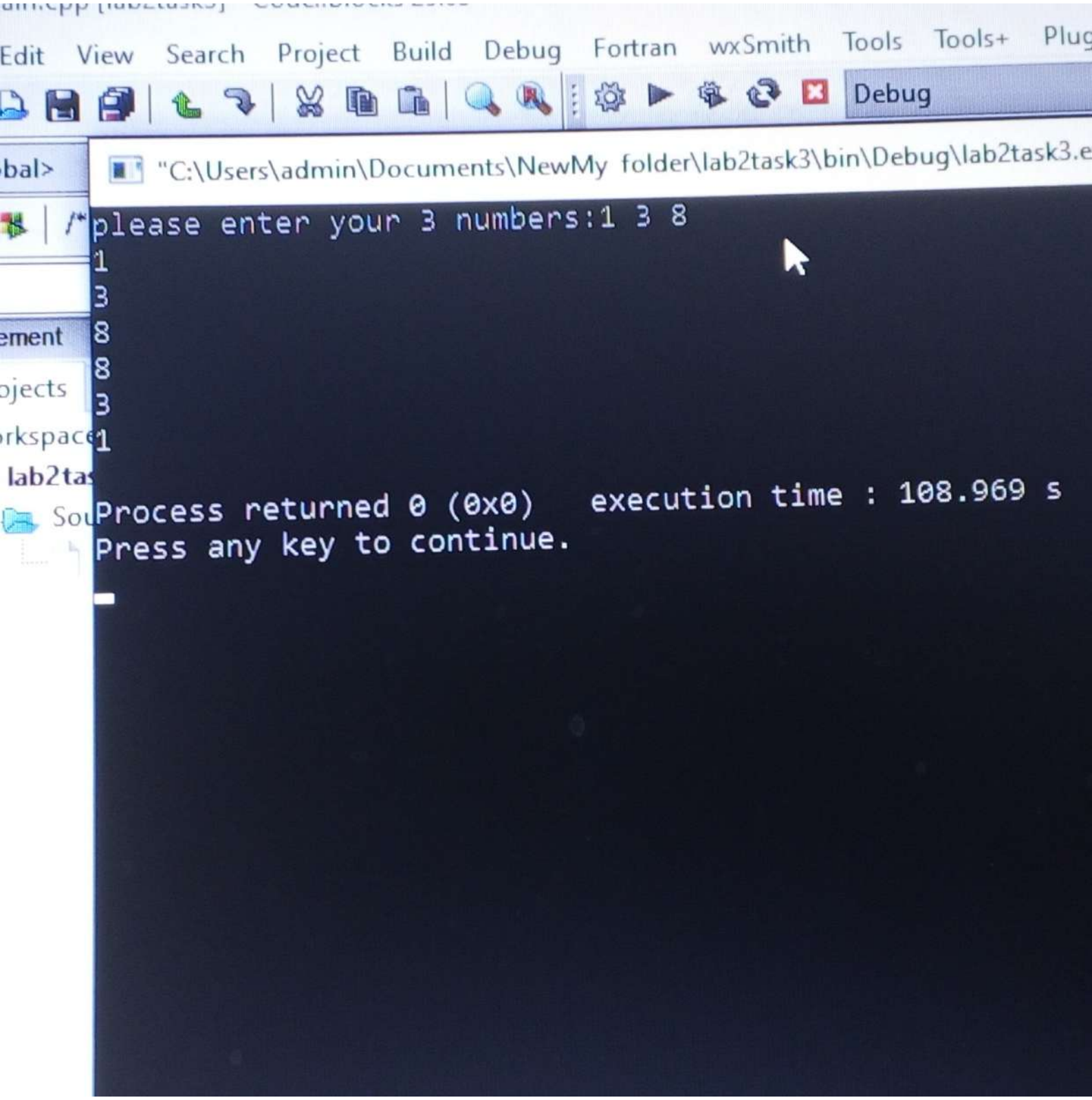

main.cp
"C:\Users\admin\Docum
12
28
20
5
Process returned 0
Press any key to con

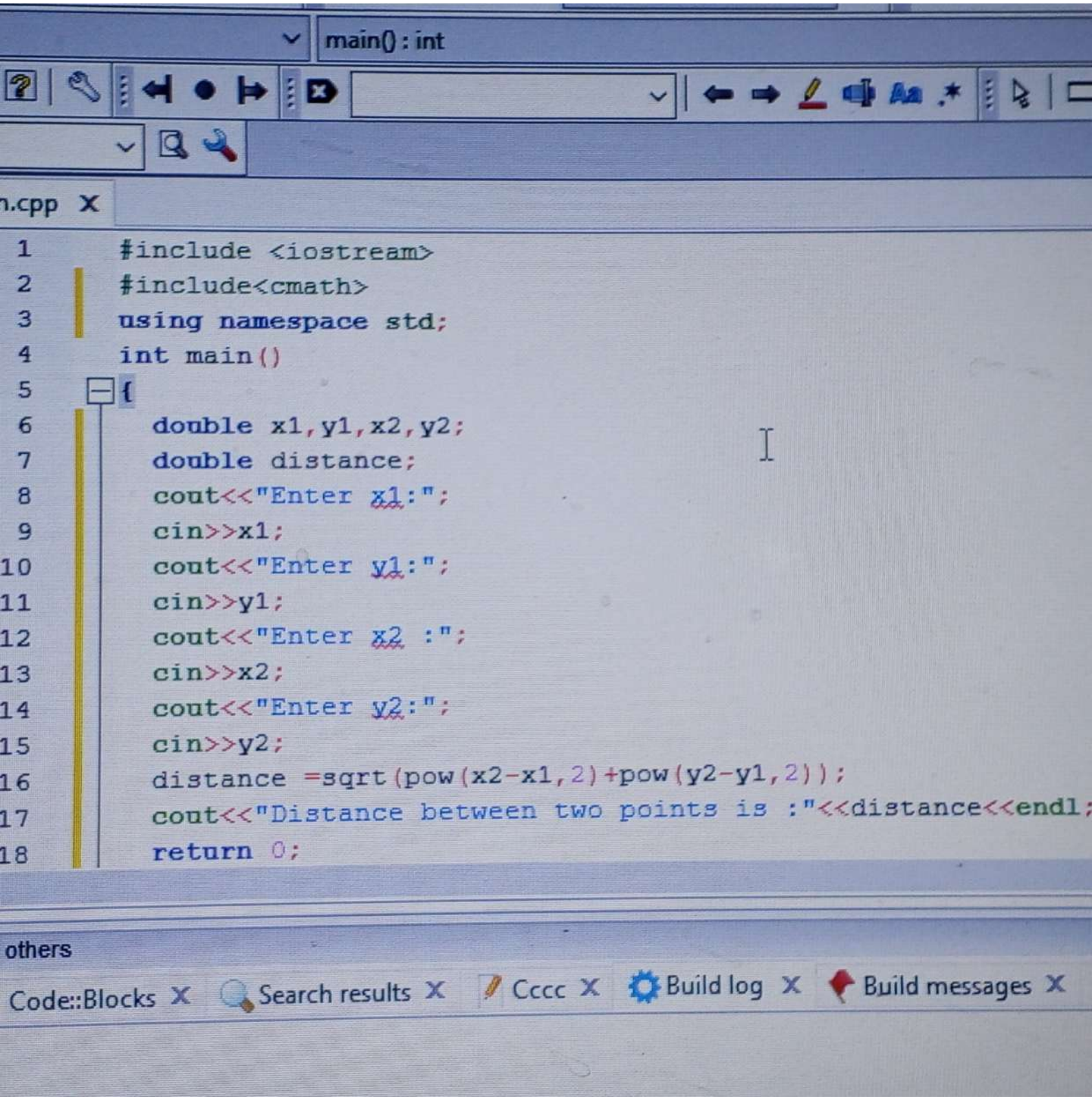
main.cpp

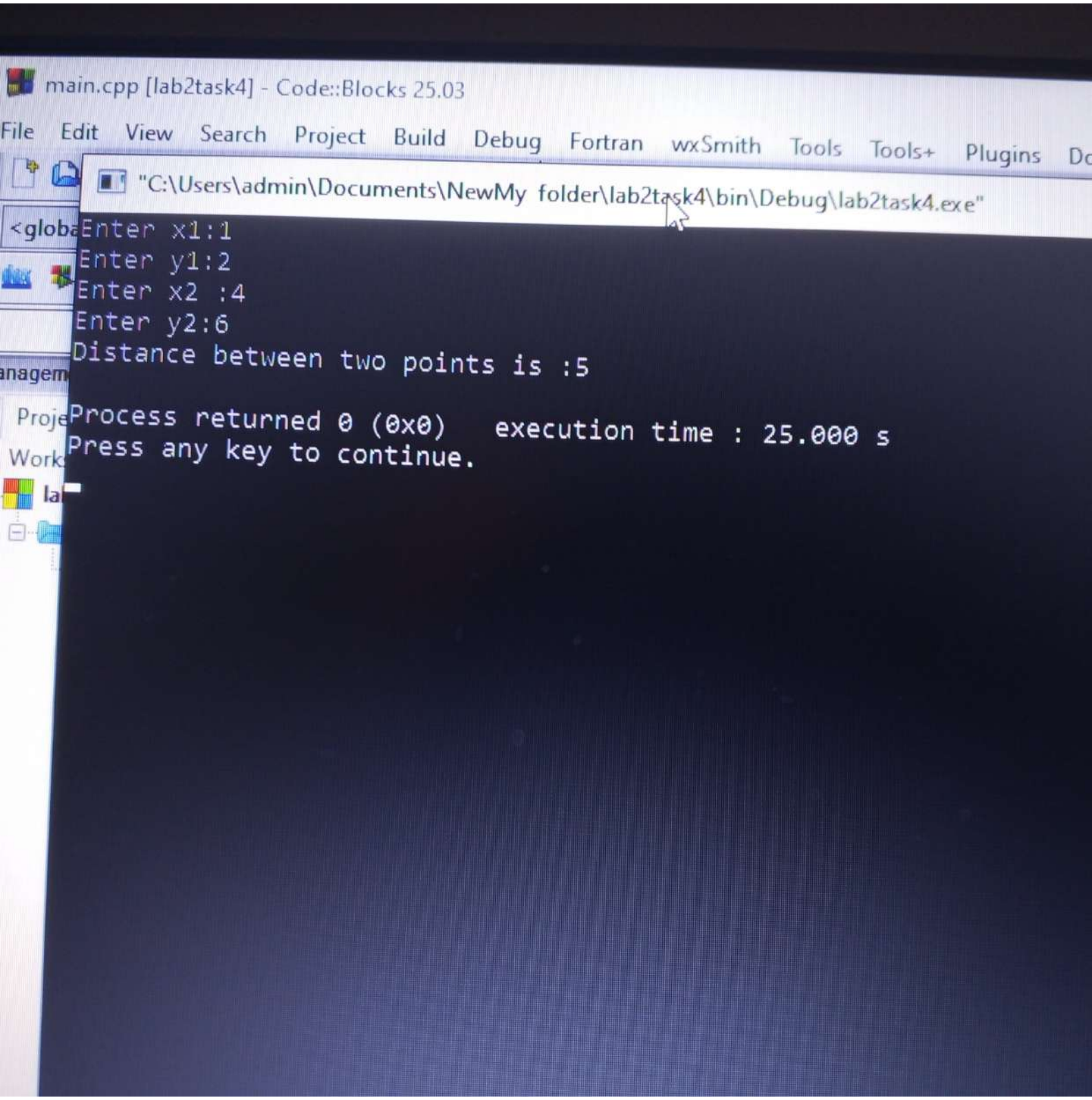
```
1   #include <iostream>
2
3   using namespace std;
4
5   int main()
6   {
7       int val1;
8       int val2;
9       int val3;
10      cout<<"please enter your 3 numbers:";
11      cin>>val1>>val2>>val3;
12      cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
13      cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
14      return 0;
15  }
16
```

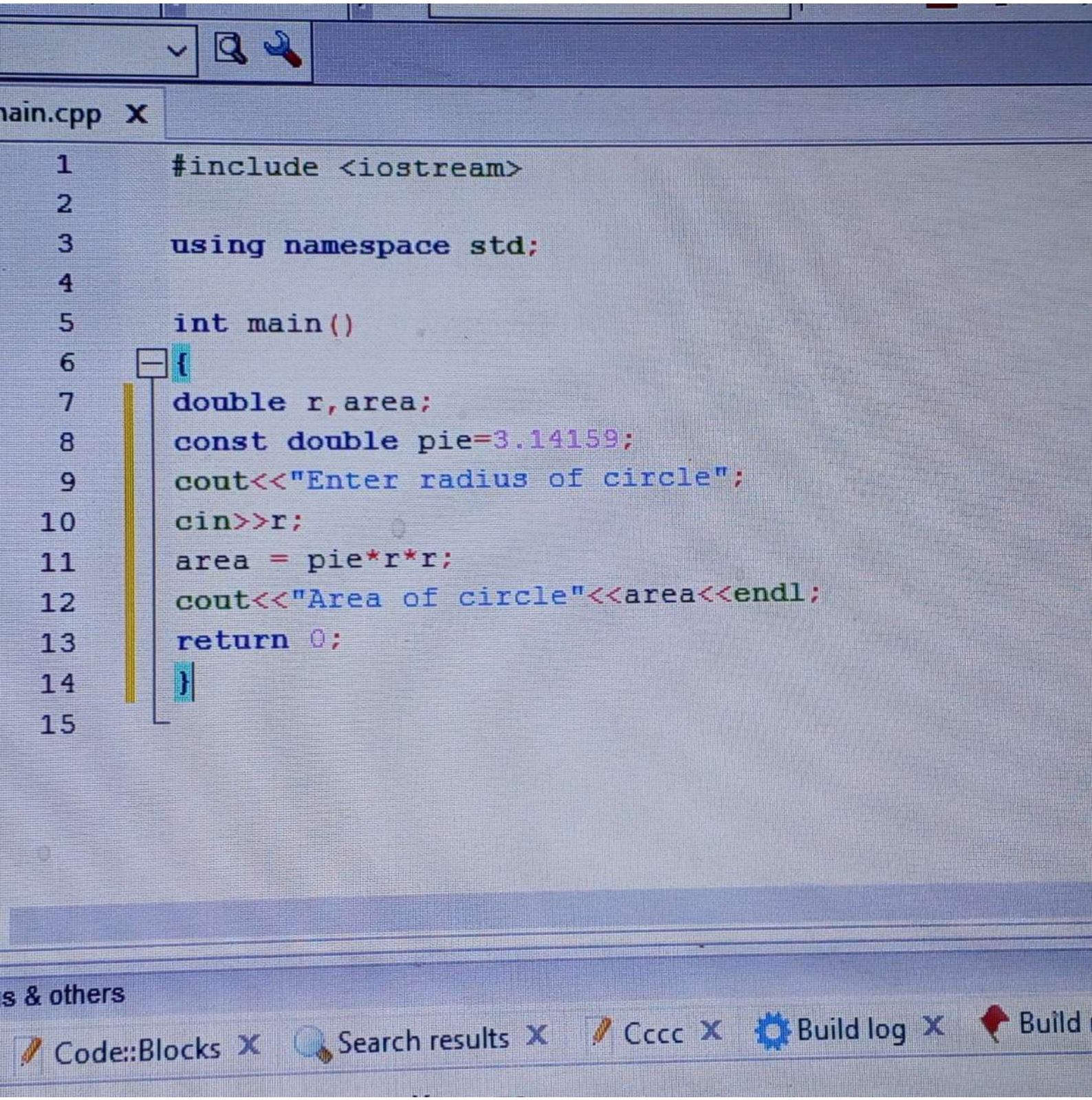
s & others

Code::Blocks | Search results | Cccc | Build log | Build messag


Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plug
Debug
"C:\Users\admin\Documents\NewMy folder\lab2task3\bin\Debug\lab2task3.e
please enter your 3 numbers:1 3 8
1
3
8
8
3
1
Process returned 0 (0x0) execution time : 108.969 s
Press any key to continue.


main() : int
n.cpp
1 #include <iostream>
2 #include<cmath>
3 using namespace std;
4 int main()
5 {
6 double x1,y1,x2,y2;
7 double distance;
8 cout<<"Enter x1:";
9 cin>>x1;
10 cout<<"Enter y1:";
11 cin>>y1;
12 cout<<"Enter x2 :";
13 cin>>x2;
14 cout<<"Enter y2:";
15 cin>>y2;
16 distance =sqrt(pow(x2-x1,2)+pow(y2-y1,2));
17 cout<<"Distance between two points is :"<<distance<<endl;
18 return 0;
others
Code::Blocks
Search results
Cccc
Build log
Build messages


main.cpp [lab2task4] - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins
"C:\Users\admin\Documents\NewMy folder\lab2task4\bin\Debug\lab2task4.exe"
Enter x1:1
Enter y1:2
Enter x2 :4
Enter y2:6
Distance between two points is :5
Process returned 0 (0x0) execution time : 25.000 s
Press any key to continue.

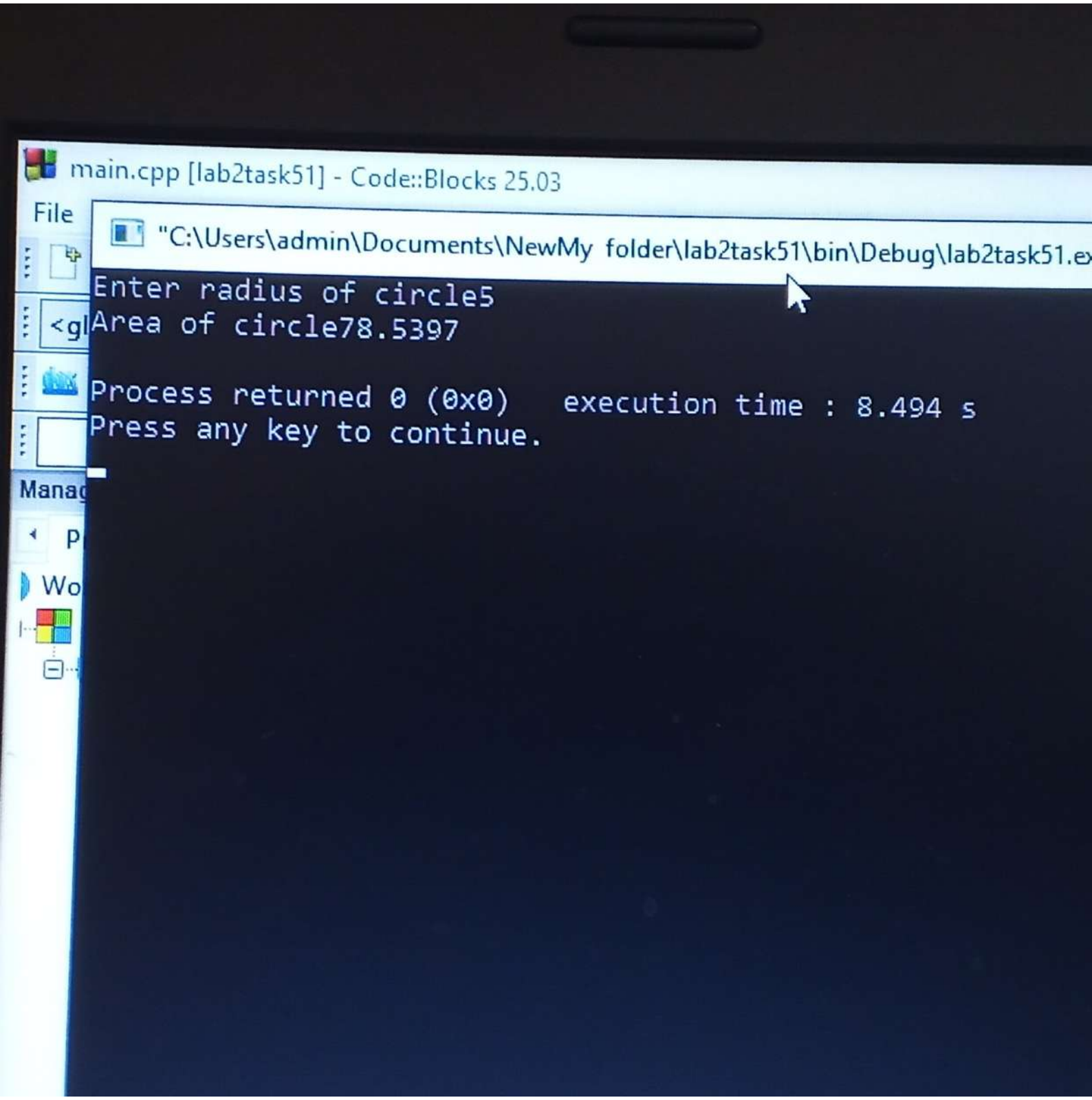


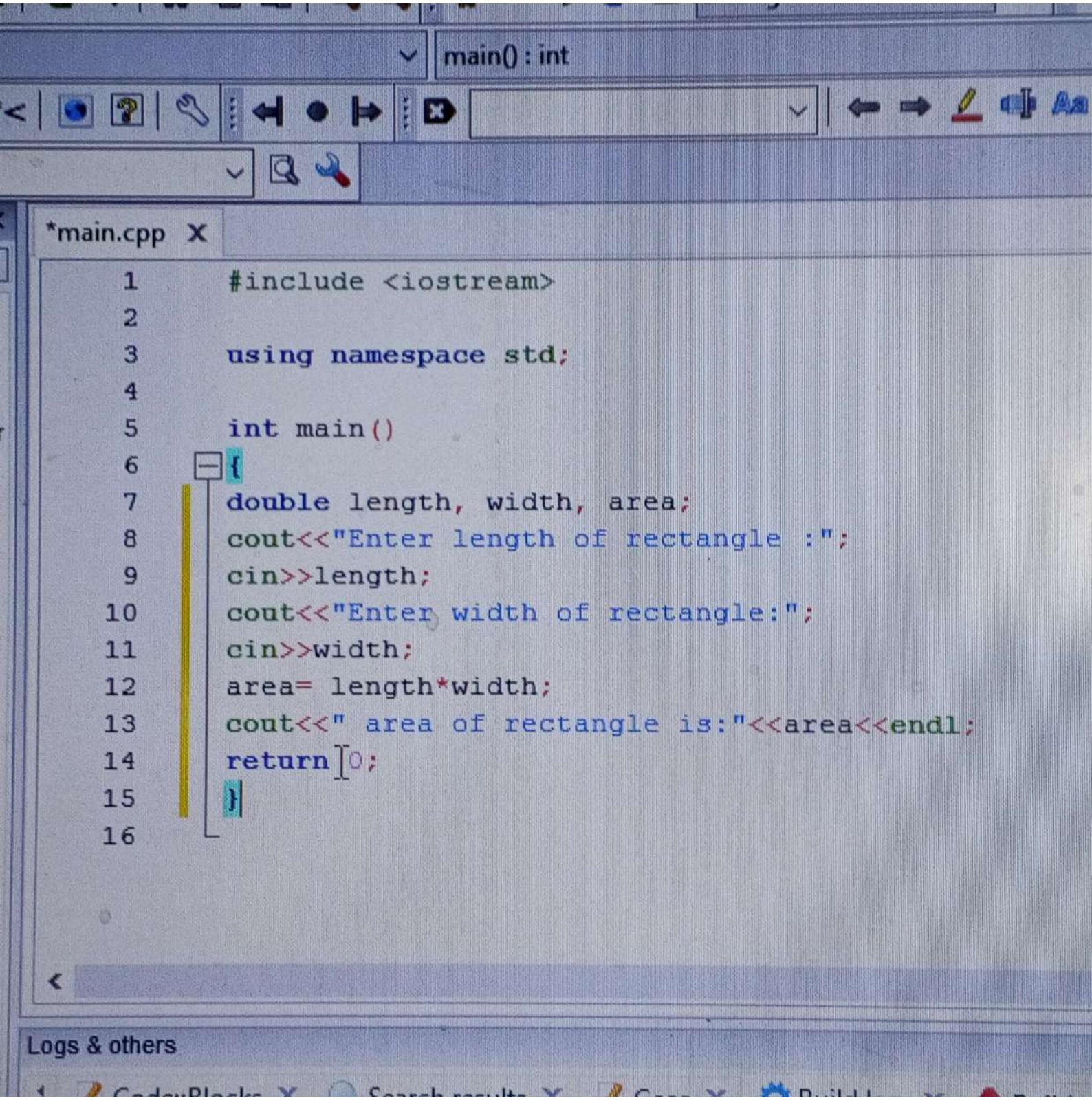
main.cpp

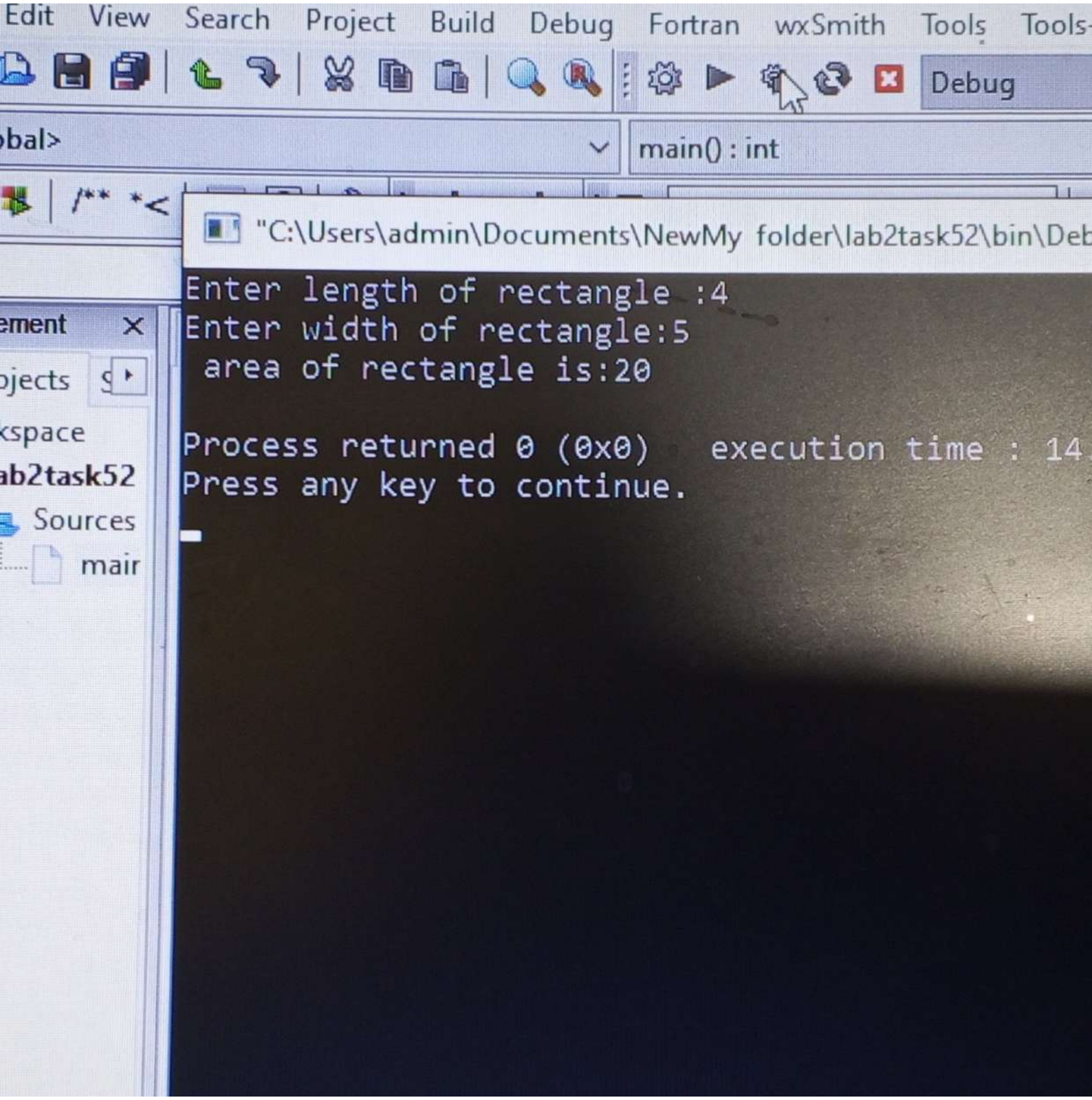
```
1   #include <iostream>
2
3   using namespace std;
4
5   int main()
6   {
7   double r,area;
8   const double pie=3.14159;
9   cout<<"Enter radius of circle";
10  cin>>r;
11  area = pie*r*r;
12  cout<<"Area of circle"<<area<<endl;
13  return 0;
14  }
15
```

s & others

Code::Blocks X | Search results X | Cccc X | Build log X | Build


main.cpp [lab2task51] - Code::Blocks 25.03
File
"C:\Users\admin\Documents\NewMy folder\lab2task51\bin\Debug\lab2task51.ex
Enter radius of circle5
Area of circle78.5397
Process returned 0 (0x0) execution time : 8.494 s
Press any key to continue.


main() : int
*main.cpp
1 #include <iostream>
2
3 using namespace std;
4
5 int main()
6 {
7 double length, width, area;
8 cout<<"Enter length of rectangle :";
9 cin>>length;
10 cout<<"Enter width of rectangle:";
11 cin>>width;
12 area= length*width;
13 cout<<" area of rectangle is:"<<area<<endl;
14 return 0;
15 }
16
Logs & others


Edit View Search Project Build Debug Fortran wxSmith Tools Tools-
Debug
main() : int
"C:\Users\admin\Documents\NewMy folder\lab2task52\bin\Deb
Enter length of rectangle :4
Enter width of rectangle:5
area of rectangle is:20
Process returned 0 (0x0) execution time : 14
Press any key to continue.
Sources
mair

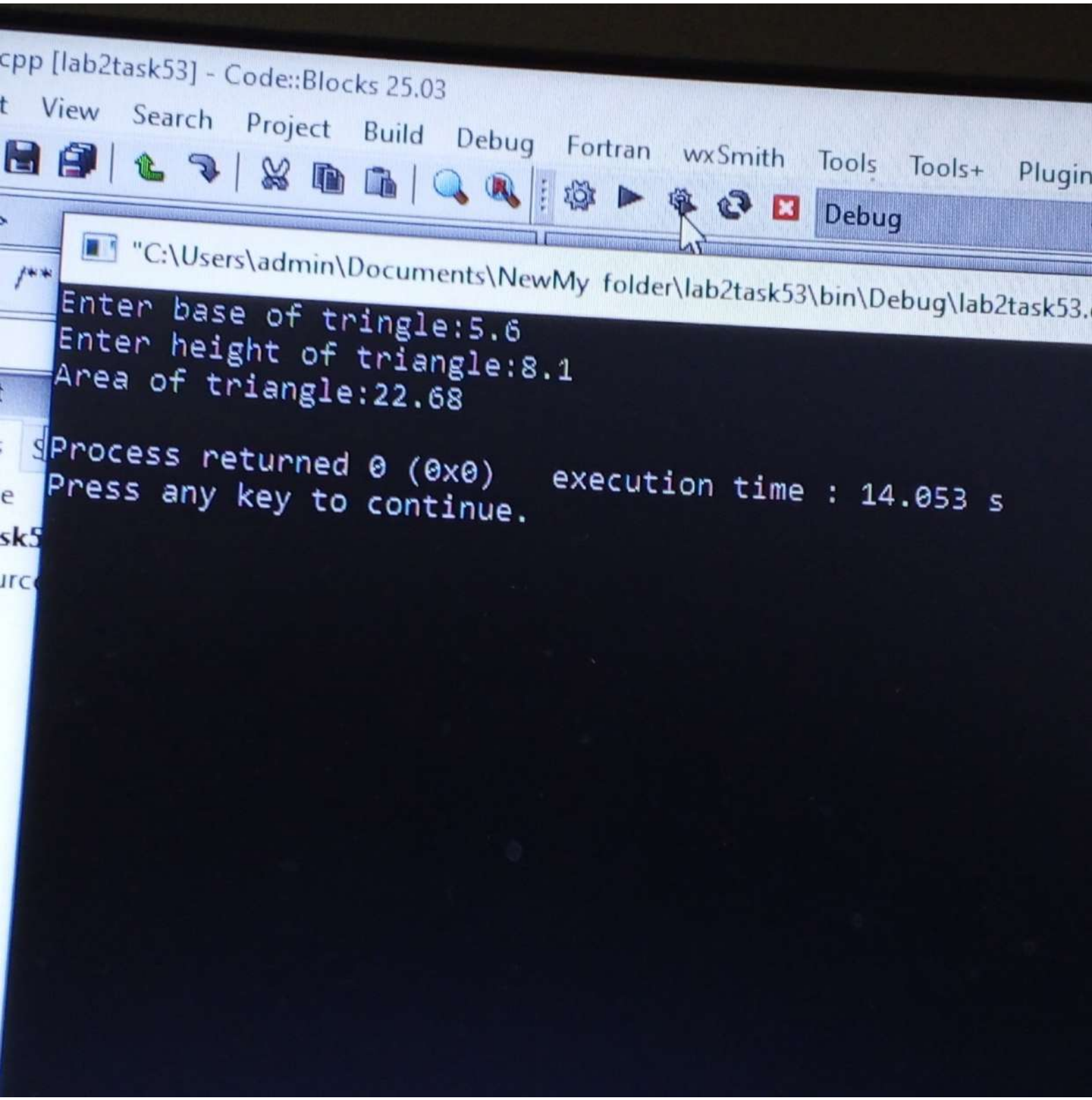
main.cpp

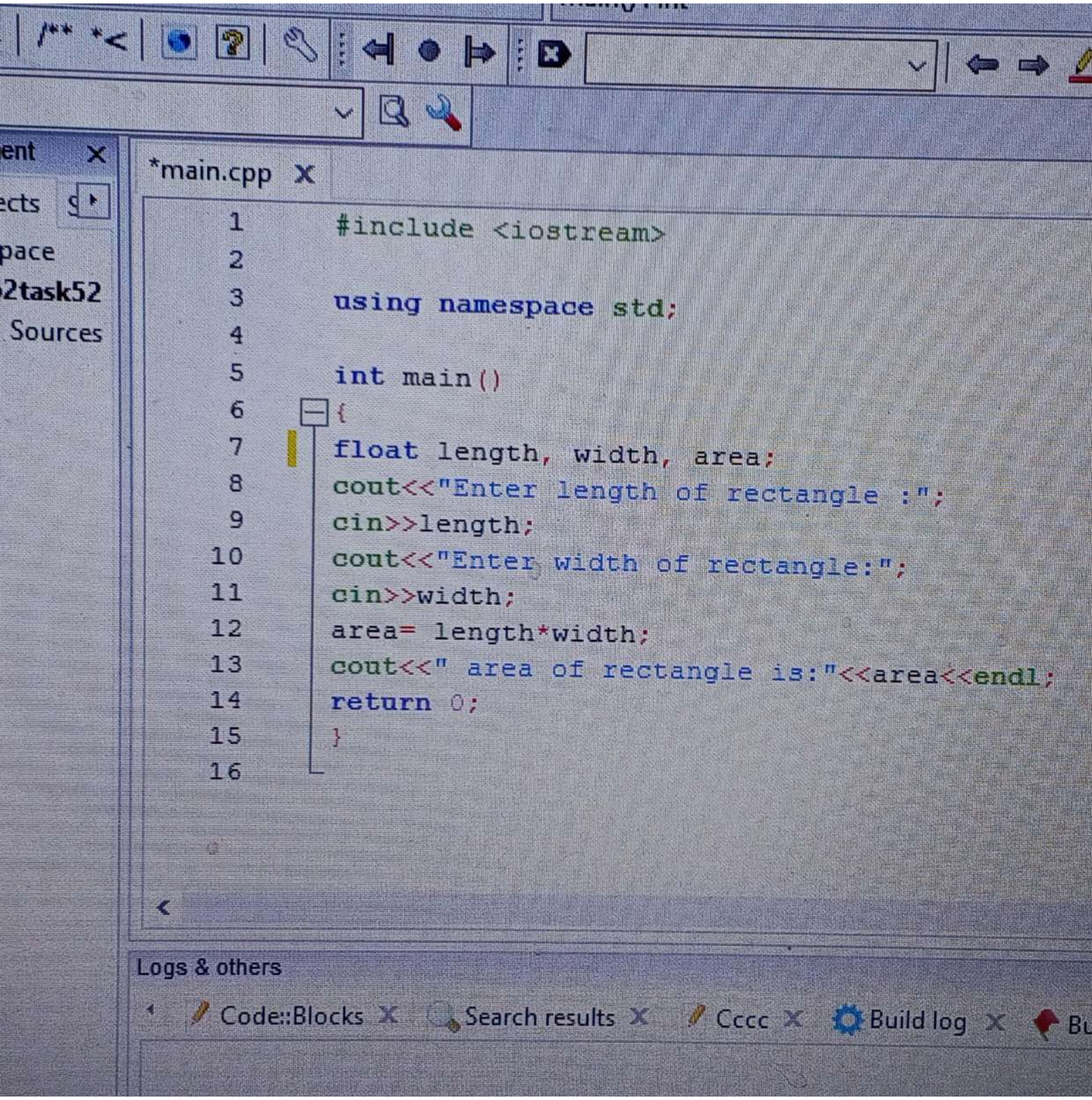
```cpp
#include <iostream>

using namespace std;

int main()
{
float base,height,area;
cout<<"Enter base of tringle:";
cin>>base;
cout<<"Enter height of triangle:";
cin>>height;
area= base*height*0.5;
cout<<"Area of triangle:"<<area<<endl;
return 0;
}
```

Logs & others


cpp [lab2task53] - Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugin
Debug
"C:\Users\admin\Documents\NewMy folder\lab2task53\bin\Debug\lab2task53.
Enter base of tringle:5.6
Enter height of triangle:8.1
Area of triangle:22.68
Process returned 0 (0x0) execution time : 14.053 s
Press any key to continue.



*main.cpp

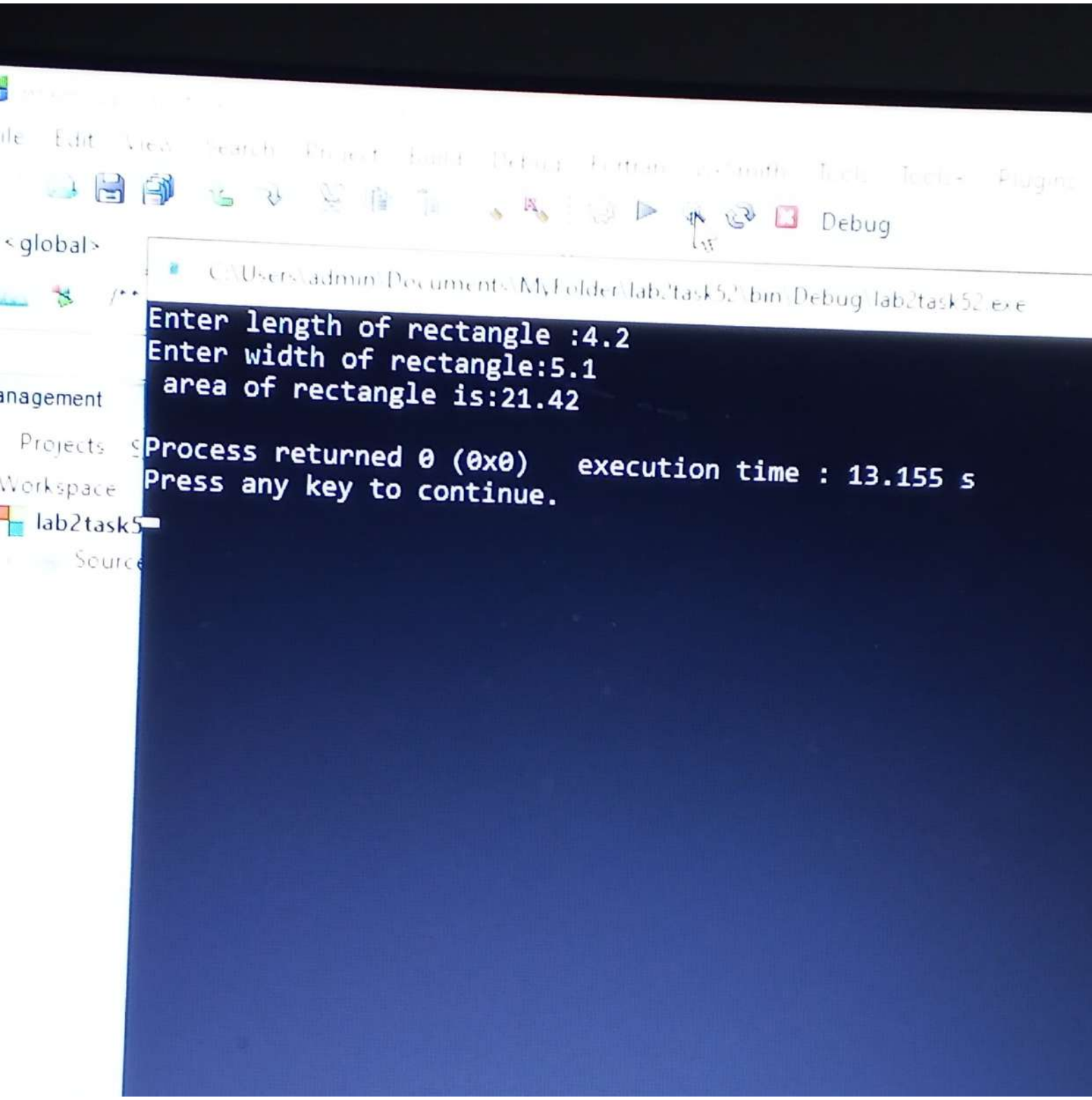
```
#include <iostream>

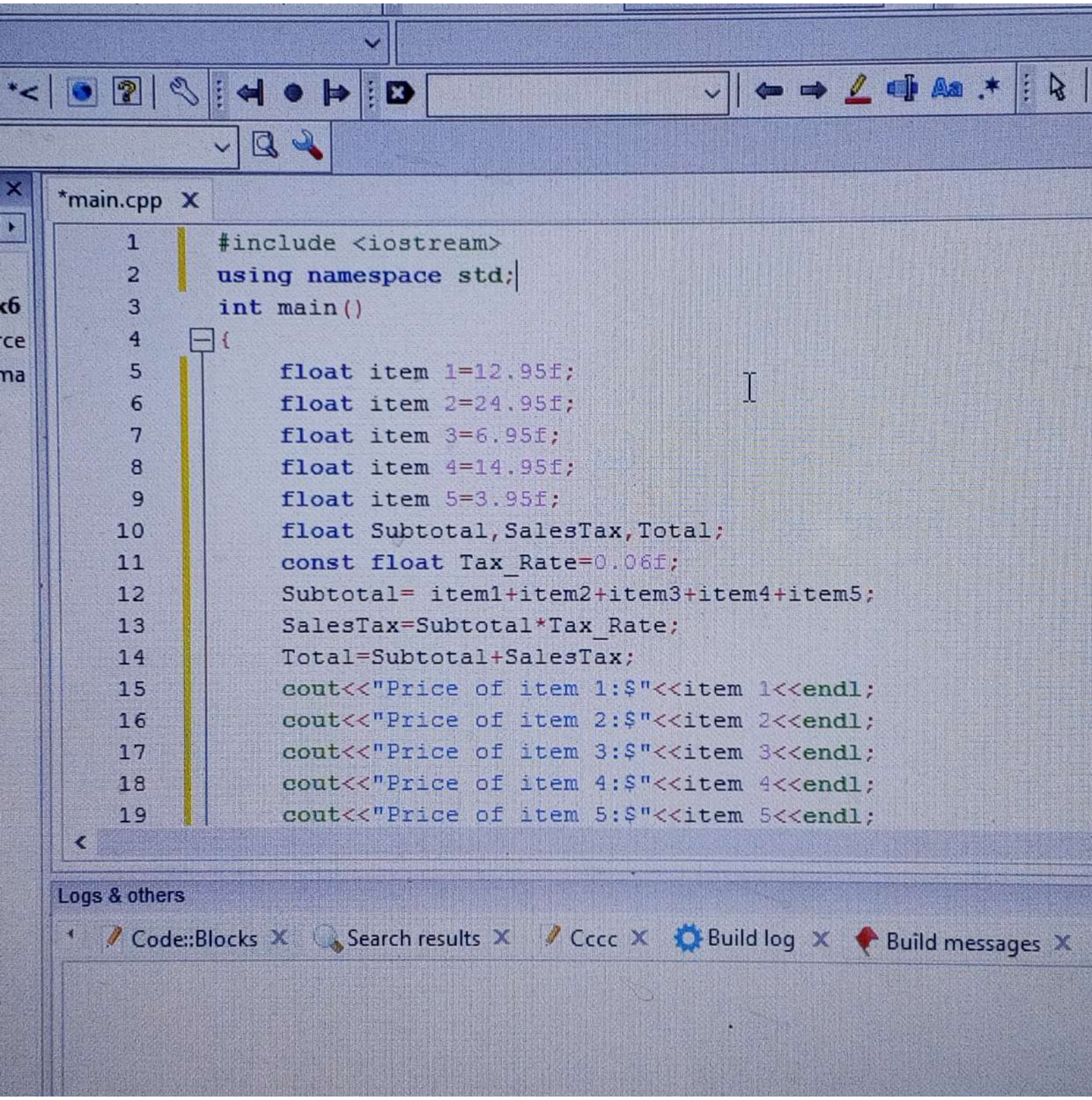
using namespace std;

int main()
{
float length, width, area;
cout<<"Enter length of rectangle :";
cin>>length;
cout<<"Enter width of rectangle:";
cin>>width;
area= length*width;
cout<<" area of rectangle is:"<<area<<endl;
return 0;
}
```

Logs & others

Code::Blocks | Search results | Cccc | Build log


Debug
<global>
Enter length of rectangle :4.2
Enter width of rectangle:5.1
area of rectangle is:21.42
Process returned 0 (0x0) execution time : 13.155 s
Press any key to continue.
lab2task5



*main.cpp

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      float item 1=12.95f;
6      float item 2=24.95f;
7      float item 3=6.95f;
8      float item 4=14.95f;
9      float item 5=3.95f;
10     float Subtotal,SalesTax,Total;
11     const float Tax_Rate=0.06f;
12     Subtotal= item1+item2+item3+item4+item5;
13     SalesTax=Subtotal*Tax_Rate;
14     Total=Subtotal+SalesTax;
15     cout<<"Price of item 1:$"<<item 1<<endl;
16     cout<<"Price of item 2:$"<<item 2<<endl;
17     cout<<"Price of item 3:$"<<item 3<<endl;
18     cout<<"Price of item 4:$"<<item 4<<endl;
19     cout<<"Price of item 5:$"<<item 5<<endl;
```

Logs & others

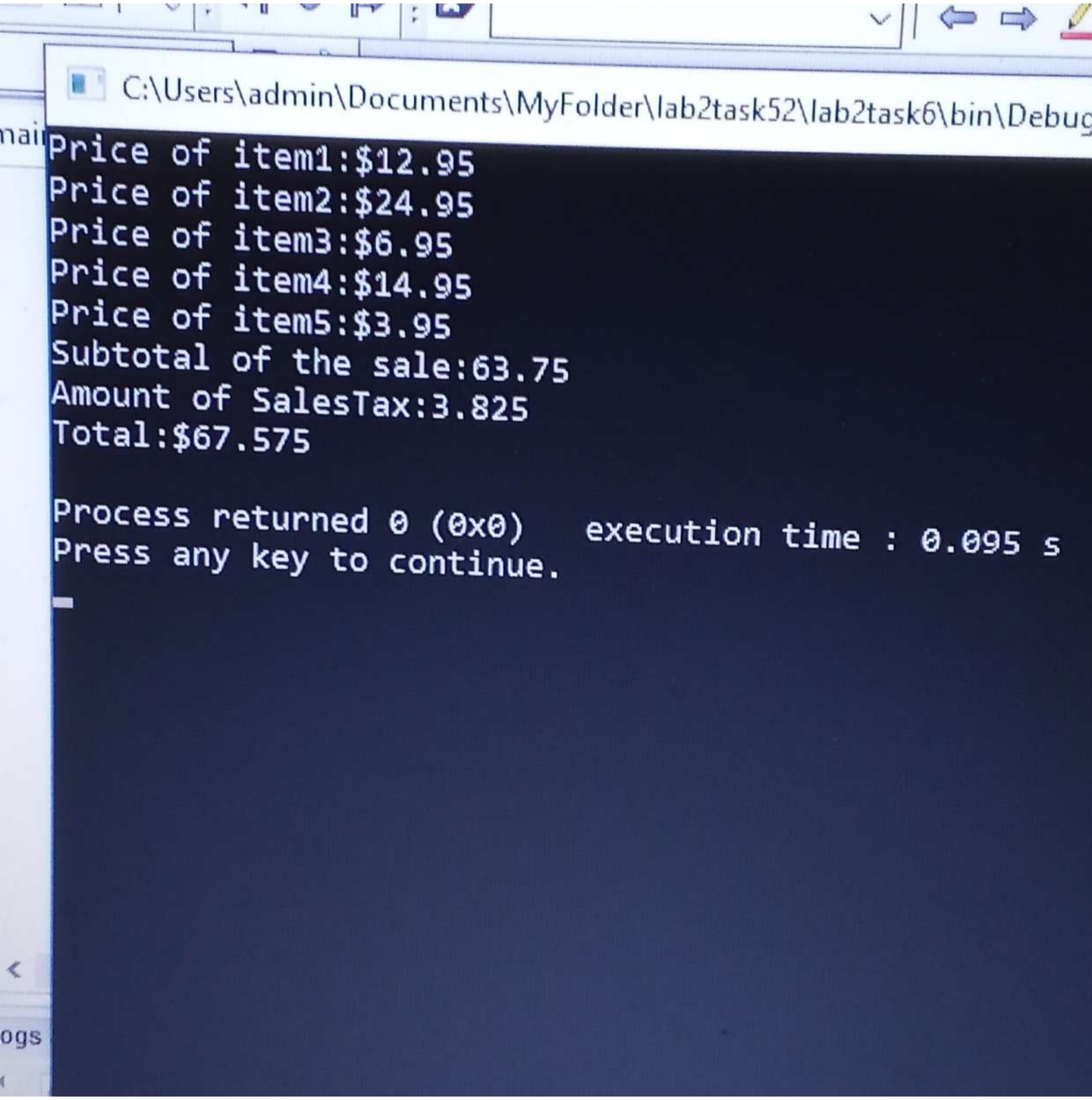
Code::Blocks | Search results | Cccc | Build log | Build messages

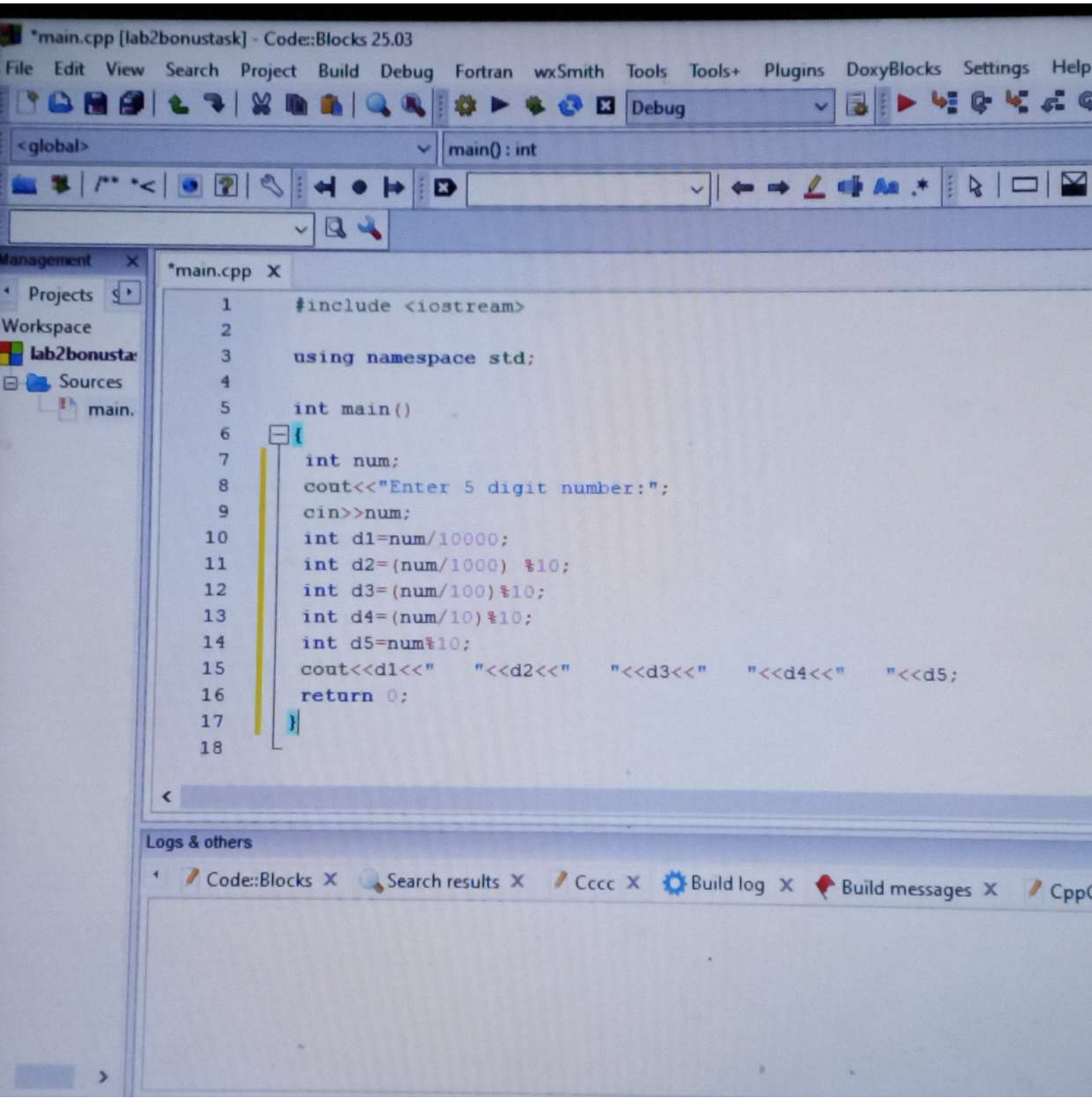
.cpp X

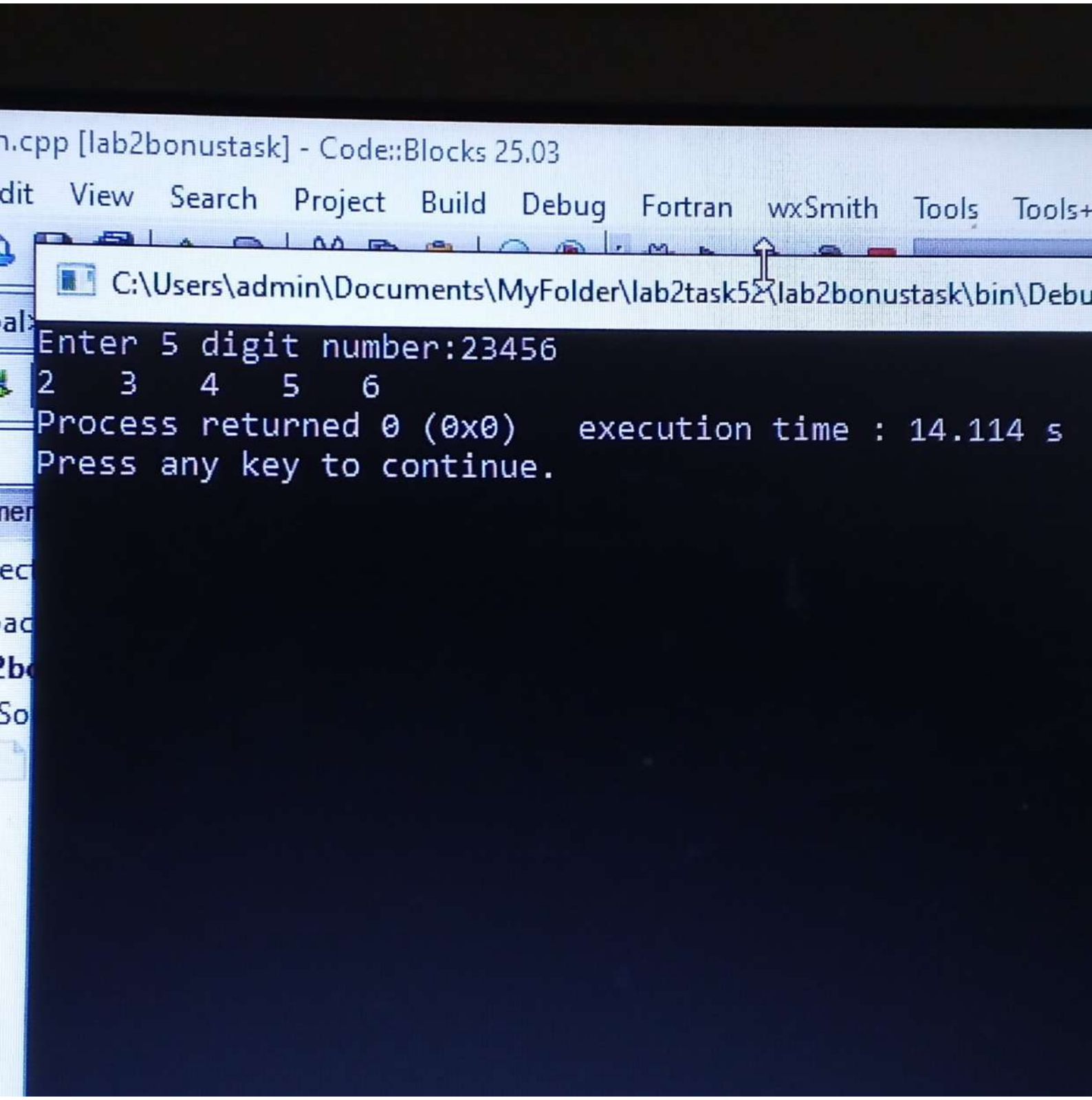
```
8       float item 4=14.95f;
9       float item 5=3.95f;
10      float Subtotal,SalesTax,Total;
11      const float Tax_Rate=0.06f;
12      Subtotal= item1+item2+item3+item4+item5;
13      SalesTax=Subtotal*Tax_Rate;
14      Total=Subtotal+SalesTax;
15      cout<<"Price of item 1:$"<<item 1<<endl;
16      cout<<"Price of item 2:$"<<item 2<<endl;
17      cout<<"Price of item 3:$"<<item 3<<endl;
18      cout<<"Price of item 4:$"<<item 4<<endl;
19      cout<<"Price of item 5:$"<<item 5<<endl;
20      cout<<"Subtotal of the sale"<<Subtotal<<endl;
21      cout<<"Amount of SaleTax"<<SaleTax<<endl;
22      cout<<Total:$"<<total<<endl;
23      return 0;
24  }
25
26
```

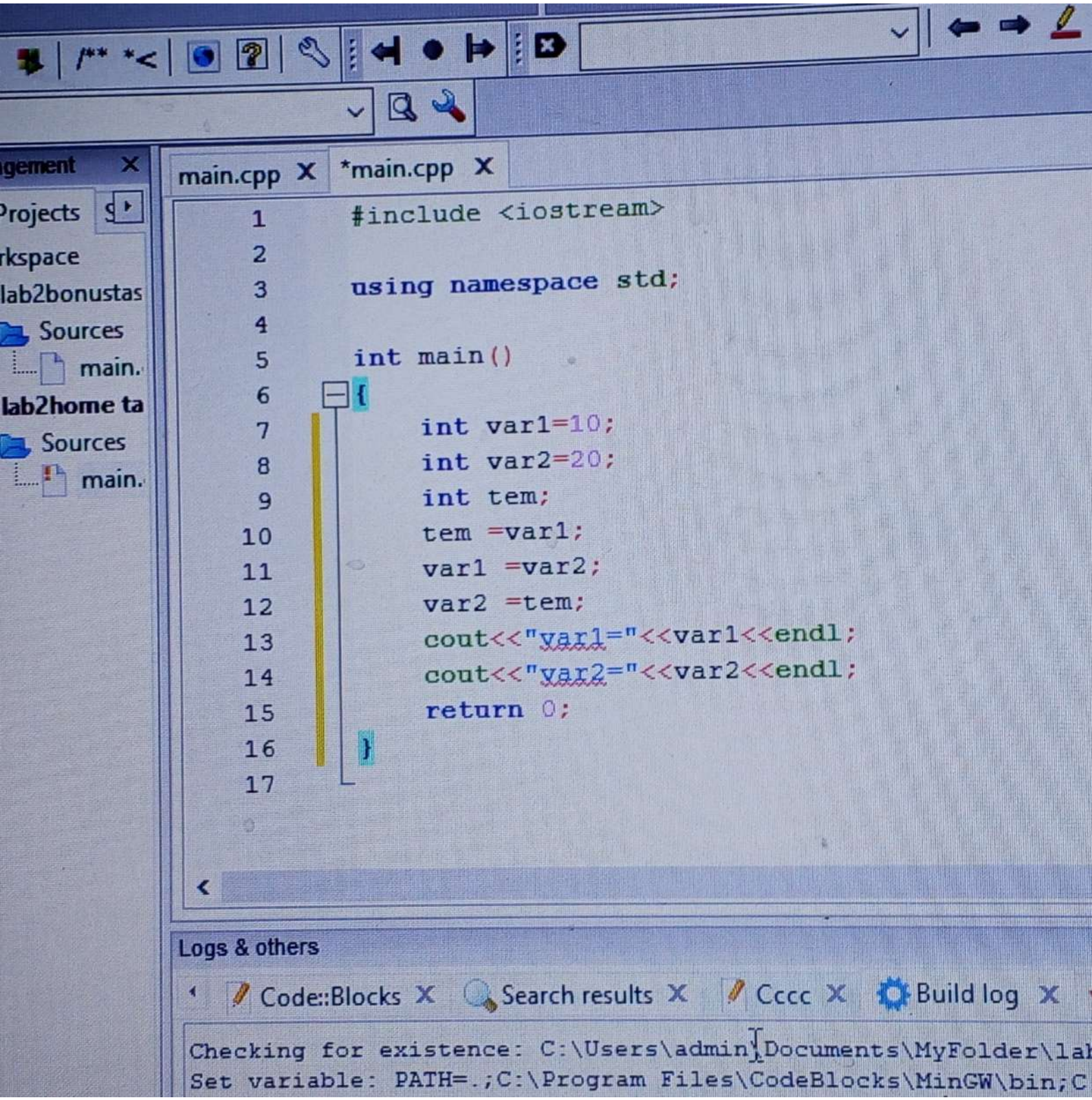
& others

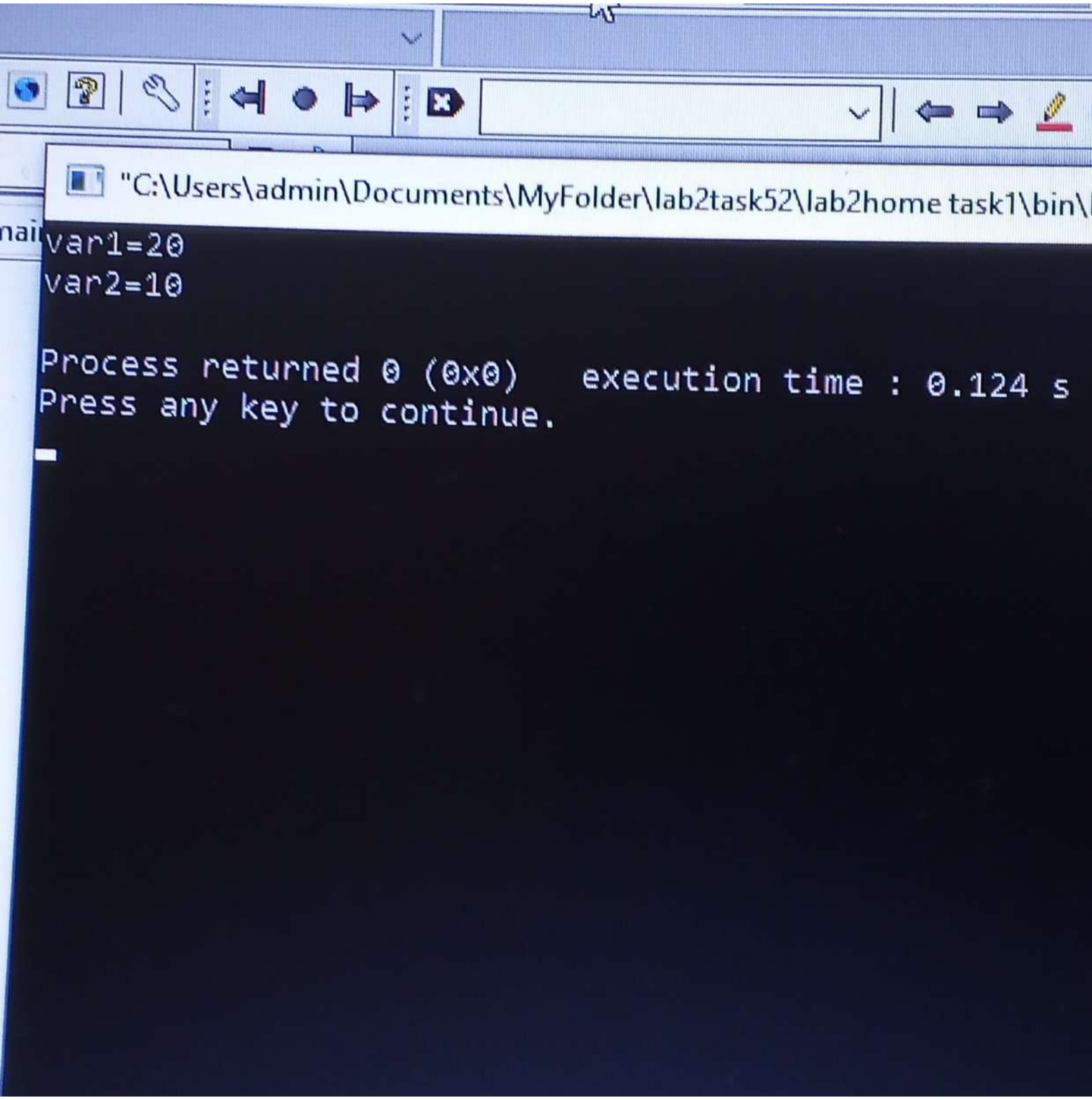
Code::Blocks X | Search results X | Cccc X | Build log X | Build messages

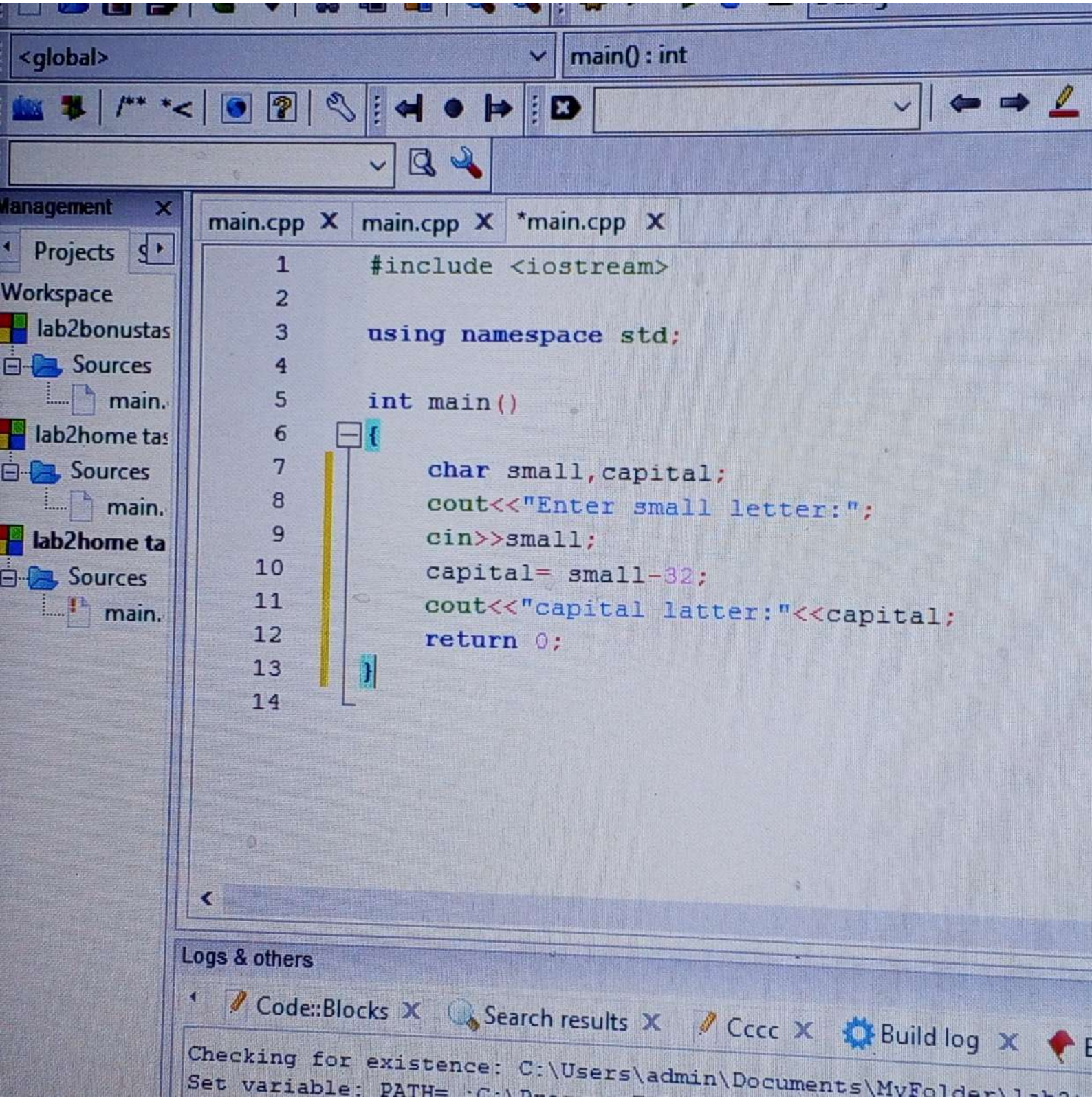

C:\Users\admin\Documents\MyFolder\lab2task52\lab2task6\bin\Debug
Price of item1:$12.95
Price of item2:$24.95
Price of item3:$6.95
Price of item4:$14.95
Price of item5:$3.95
Subtotal of the sale:63.75
Amount of SalesTax:3.825
Total:$67.575
Process returned 0 (0x0)   execution time : 0.095 s
Press any key to continue.

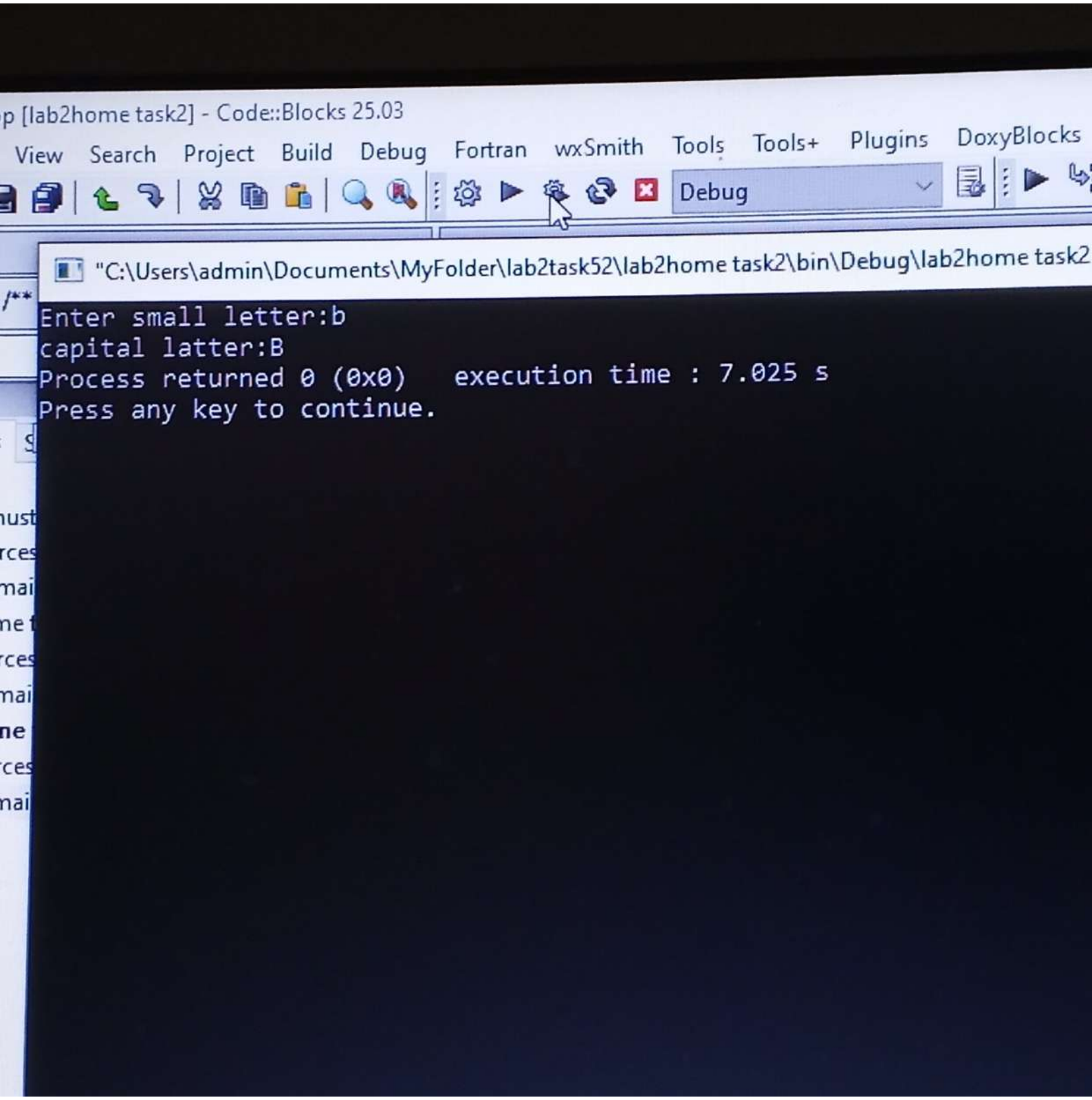

*main.cpp [lab2bonustask] - Code::Blocks 25.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Debug
<global>
main() : int
Management
Projects
Workspace
lab2bonusta
Sources
main.
*main.cpp
1 #include <iostream>
2
3 using namespace std;
4
5 int main()
6 {
7 int num;
8 cout<<"Enter 5 digit number:";
9 cin>>num;
10 int d1=num/10000;
11 int d2=(num/1000) %10;
12 int d3=(num/100)%10;
13 int d4=(num/10)%10;
14 int d5=num%10;
15 cout<<d1<<" "<<d2<<" "<<d3<<" "<<d4<<" "<<d5;
16 return 0;
17 }
18
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages


n.cpp [lab2bonustask] - Code::Blocks 25.03
dit View Search Project Build Debug Fortran wxSmith Tools Tools+
C:\Users\admin\Documents\MyFolder\lab2task52\lab2bonustask\bin\Debu
Enter 5 digit number:23456
2 3 4 5 6
Process returned 0 (0x0) execution time : 14.114 s
Press any key to continue.


main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main()
{
int var1=10;
int var2=20;
int tem;
tem =var1;
var1 =var2;
var2 =tem;
cout<<"var1="<<var1<<endl;
cout<<"var2="<<var2<<endl;
return 0;
}
Logs & others
Code::Blocks
Search results
Cccc
Build log
Checking for existence: C:\Users\admin\Documents\MyFolder\lab
Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\bin;C


"C:\Users\admin\Documents\MyFolder\lab2task52\lab2home task1\bin\
var1=20
var2=10
Process returned 0 (0x0)   execution time : 0.124 s
Press any key to continue.


<global>
main() : int
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Sources
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main.cpp
main.cpp
*main.cpp
1 #include <iostream>
2
3 using namespace std;
4
5 int main()
6 {
7 char small,capital;
8 cout<<"Enter small letter:";
9 cin>>small;
10 capital= small-32;
11 cout<<"capital latter:"<<capital;
12 return 0;
13 }
14
Logs & others
Code::Blocks
Search results
Cccc
Build log
Checking for existence: C:\Users\admin\Documents\MyFolder
Set variable: PATH=


p [lab2home task2] - Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks
Debug
"C:\Users\admin\Documents\MyFolder\lab2task52\lab2home task2\bin\Debug\lab2home task2
Enter small letter:b
capital latter:B
Process returned 0 (0x0) execution time : 7.025 s
Press any key to continue.